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tactile-to-visual representation learning for vision-language-action models

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~~Cross-modal~~ ~~Tactile~~ ~~OpenVLA~~ ~~Frozen~~ ~~Latent~~ ~~Vision~~ ~~Action~~ ~~Language~~ ~~Sensor~~ ~~Robot~~

A mamma, per essere stata la mia
finestra sul mondo. E per esserlo
tutt'oggi, quando, nonostante la mia
statura, non arrivo ancora al davanzale.
A Sofia, giudice mai clemente,
per riportarmi con i piedi a terra
e sostenermi senza esitazioni.
Ai miei amici.
A Francesco, Stefan, Riccardo e Andrea,
compagni di un tempo che non tornerà,
per ciò che assieme abbiamo condiviso
ed affrontato.
A Yuri, per aver saputo ascoltare
e comprendere anche i miei silenzi.
A Célia, musa onnipresente, per mostrarmi
le sfumature d'ogni colore.
Per essere il mio scorcio di libertà.
Per aver dato forma all'amore.
Guarderò sempre nella tua direzione,
anche quando mi sarà difficile vedere
quello che vedi tu.
You must be the place.

Abstract

Vision-Language-Action (VLA) models have demonstrated strong generalisation capabilities in robotic manipulation, yet their action generation relies primarily on visual input, which is often degraded or entirely unavailable during contact-rich interactions such as grasping and in-hand manipulation. This thesis investigates whether a frozen VLA can maintain task performance under degraded or absent vision by learning a lightweight projection module, f_θ, that maps tactile embeddings into the policy's frozen visual latent space without modifying the pretrained policy weights. Rather than committing immediately to full-scale training, the work follows a feasibility-first methodology designed to establish whether such an approach is well-founded. First, a representational similarity analysis of tactile and visual embedding spaces is conducted on a small exploratory dataset, leading to the evidence-based selection of a generic ViT-Small tactile backbone over the official TVL encoder. Second, a fully reproducible benchmark is developed by integrating a custom DIGIT-equipped Panda gripper into the LIBERO/robosuite/MuJoCo simulation stack, extending TACTO's tactile rendering pipeline to a new simulator. This integration also reveals an independent mechanical confound: the gripper modification alone reduces the frozen policy's success rate on `libero_object` from 98.2% to 56.8%, before any tactile alignment is introduced. Third, the projection module is trained and evaluated under a leave-one-object-out protocol against a naive baseline. Although it achieves a mean cross-object cosine similarity of 0.852, it does not consistently outperform the baseline, providing inconclusive evidence of cross-object generalisation at the scale of the available data. Taken together, these findings support a qualified rather than unqualified answer to the central hypothesis of this thesis. Tactile and visual embeddings appear sufficiently compatible, in aggregate, to justify learning a shared representation, yet generalisation to unseen objects remains unproven, while an independent mechanical confound limits policy performance regardless of the alignment strategy. Overall, the principal contribution of this thesis is a validated experimental infrastructure and a rigorous feasibility study, rather than a demonstrated end-to-end capability.

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1

Introduction

1.1 Motivation and Problem Statement

Vision-Language-Action (VLA) models have emerged as the dominant paradigm for generalist robotic manipulation, combining the semantic priors of large pretrained vision-language backbones with the ability to generate low-level robot actions directly from raw sensory input and natural-language instructions [1], [2], [3]. Their generalisation capability stems almost entirely from the visual modality: actions are conditioned on a latent representation produced by a vision encoder that has been pretrained on internet-scale image-text data and subsequently exposed to large-scale robot demonstration datasets.

This dependence on vision is also the paradigm's most significant weakness. In real-world manipulation, particularly during contact-rich phases such as grasping, insertion or in-hand manipulation, visual observation is frequently degraded or entirely unavailable: the end-effector or the manipulated object may occlude the camera, ambient lighting may be poor or unstable and fine-grained contact geometry is, by construction, invisible to a camera positioned outside the contact region. Humans compensate for exactly this limitation using tactile feedback, which becomes the dominant sensory channel precisely when vision is least informative. Equipping VLA policies with an analogous capability would allow them to degrade gracefully rather than catastrophically when visual input is compromised.

A naive solution would be to retrain or fine-tune a VLA end-to-end on multimodal visuo-tactile data. This approach, however, is computationally expensive, requires large-scale paired demonstrations and risks catastrophic forgetting of the visual-semantic priors that make the pretrained backbone useful in the first place [3]. An alternative, motivated by recent evidence that pretrained VLMs already encode substantial latent knowledge about physical interaction [4], is to leave the VLA backbone completely frozen and instead learn a lightweight external module that translates tactile signals into the *same* latent space the frozen vision encoder would have produced. If such a projection is sufficiently accurate in replicating the original visual embedding, the frozen policy can, in principle, continue to generate coherent actions even when its visual input is corrupted or missing, without any modification to its pretrained weights.

This thesis investigates the feasibility of exactly this idea: a tactile-to-visual projection module that allows a frozen VLA to maintain task performance under degraded or absent visual conditions, by dynamically fusing tactile-derived representations into the policy's existing visual latent space.

1.2 Research Question and Hypothesis

The central research question addressed by this thesis is the following:

> *Can a frozen Vision-Language-Action model maintain task performance under degraded or absent visual input by dynamically fusing tactile-derived representations into its visual latent space, without any modification to the base policy weights?*

The underlying hypothesis is that the visual latent space of a pretrained VLA is sufficiently general to accommodate representations derived from a different sensory modality and that a lightweight learned projection $f_\theta : z_\mathcal{T} \to z_\mathcal{V}$ can produce tactile embeddings that are behaviourally equivalent to visual ones when vision is unreliable. Under nominal visual conditions, the system is expected to recover the original behaviour of the frozen VLA. As visual quality deteriorates, the tactile branch should progressively compensate, preserving task performance rather than causing an abrupt failure.

This hypothesis draws on two independent lines of evidence from the literature reviewed in Chapter 3. First, contrastive alignment between tactile and visual-linguistic embedding spaces has already been shown to be learnable from a comparatively small amount of paired data [5]. Second, pretrained vision-language backbones appear to already encode latent knowledge of physical contact concepts [4]. Taken together, these results suggest that the semantic structure of a frozen VLA's visual space is compatible, at least in principle, with tactile-derived representations and that the policy's action generation capability need not be retrained to accommodate them.

1.3 Contributions

This thesis makes the following contributions:

1. **A structured literature review.** A historical excursus spanning cross-modal visuo-tactile perception, cross-modal representation alignment and the evolution of VLA models is presented, together with an in-depth analysis of the five most directly relevant recent works, situating the proposed approach with respect to the current state of the art (Chapter 3).

2. **A tactile-to-visual projection architecture.** A lightweight, trainable projection module is proposed to map image-based tactile embeddings into the visual latent space of a frozen VLA, together with a hybrid alignment objective (Section 7.6).

3. **A simulation and data-generation pipeline.** A reproducible pipeline is developed for collecting paired tactile-visual data, integrating the DIGIT tactile sensor [6], the TACTO photorealistic tactile simulator [7] and the LIBERO manipulation benchmark [8], together with a frozen OpenVLA-OFT policy as the target backbone.

4. **Preliminary empirical analysis.** An exploratory comparison between visual and tactile embedding distributions is conducted to assess the feasibility of cross-modal alignment prior to full-scale training (Chapter 5).

For the sake of transparency, it should be noted that the scope of this thesis is limited by the time available for the project. Rather than reporting complete end-to-end policy evaluation results, this work delivers a fully configured and validated experimental environment, LIBERO combined with a frozen OpenVLA-OFT backbone and a TACTO-simulated DIGIT sensor, together with the architectural design and preliminary feasibility analysis needed to carry out the full training and evaluation described in Chapter 7. The completion of policy training and quantitative evaluation is left as the immediate next step and is discussed in Chapter 9.

1.4 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 introduces the technical background necessary to contextualise the proposed approach: Vision-Language-Action models, image-based tactile sensing and the DIGIT sensor, the TACTO simulation framework, cross-modal representation alignment, the OpenVLA and OpenVLA-OFT architectures and the LIBERO benchmark. Chapter 3 reviews the related literature, first through a historical excursus covering cross-modal visuo-tactile perception, representation alignment, the evolution of VLA models and then through an in-depth analysis of five directly relevant recent works. Chapter 4 formalises the research objective, hypothesis and scope of the thesis. Chapter 7 presents the proposed tactile-to-visual projection architecture, the fusion mechanism and the associated training objectives and strategy. Chapter 6 describes the simulation environment, benchmark configuration and data acquisition pipeline. Chapter 5 reports a preliminary exploratory analysis of tactile and visual embedding distributions. Finally, Chapter 9 discusses the results obtained, the limitations of the current work and directions for future research.

2

Background

This chapter introduces the technical foundations upon which the proposed approach is built: Vision-Language-Action models, image-based tactile sensing and the DIGIT sensor, the TACTO tactile simulation framework, cross-modal representation alignment, the OpenVLA and OpenVLA-OFT architectures and the LIBERO benchmark. The objective is to establish the terminology and conceptual grounding required to contextualise the methodology introduced in Chapter 4.

2.1 Vision-Language-Action Models

Vision-Language-Action (VLA) models are a class of generalist robotic policies that condition action generation jointly on visual observations and natural-language instructions. They emerged from the insight that large pretrained vision-language models (VLMs) [9] already encode rich semantic knowledge about objects, spatial relationships and physical interactions. By fine-tuning or extending such models with an action head, it becomes possible to learn policies that can follow open-vocabulary commands and generalise to unseen objects and scenes without task-specific retraining.

The paradigm was established by RT-1 [10], which demonstrated that a single transformer-based policy, trained on a large and diverse corpus of real-world robot demonstrations, could generalise across tasks and objects far better than prior task-specific approaches. RT-2 [1] extended this idea by casting action tokens as text tokens and fine-tuning a pretrained VLM backbone directly on robot data, thereby transferring web-scale semantic knowledge to robotic control. More recent work such as π_0 [2] replaces autoregressive token prediction with a diffusion-based action expert, while OpenVLA [3] provides fully open-source checkpoints that substantially lower the barrier to academic research.

The key property that motivates this thesis is that VLA models process visual input through a vision encoder that maps images into a compact latent space; actions are then predicted from that latent representation together with the language embedding. This decoupling between perception and action generation is precisely what makes it possible, in principle, to substitute or supplement the visual encoder's output with a representation derived from a different sensory modality, provided the two representations are sufficiently aligned.

From a training perspective, VLA models are typically trained via imitation learning on large-scale robot demonstration datasets, where the policy learns to predict expert actions given paired observations and language instructions. This supervised training signal,

combined with the rich semantic priors of the pretrained vision-language backbone, allows VLA models to generalise across tasks and embodiments without task-specific reward engineering. For this thesis, the key implication is that action generation is entirely mediated by the latent representation produced by the vision encoder: any module that can faithfully reproduce that representation from a different sensory modality can, in principle, drive the policy without modification.

2.2 Image-Based Tactile Sensing and the DIGIT Sensor

Tactile sensors measure the mechanical properties of physical contact (normal and shear forces, surface geometry, texture and compliance) and are therefore particularly informative during the phases of a manipulation task where vision is occluded or uninformative, such as grasping and in-hand manipulation.

Among the many available hardware technologies, image-based tactile sensors have become the dominant sensing paradigm in robotic manipulation research because they encode contact information as standard RGB images, making them directly compatible with the computer-vision processing pipelines already used for visual observations. The operating principle is shared across this sensor family: an elastomeric gel pad is pressed against the contact surface, deforming locally to conform to the object's geometry; an internal RGB camera, illuminated by coloured LEDs, captures the resulting deformation pattern and produces a spatially resolved image of the contact patch.

DIGIT [6] is a compact, low-cost, fingertip-sized optical tactile sensor developed at Meta AI Research. It is based on the same image-formation principle described above, with an open-source hardware design that makes it straightforwardly integratable on standard parallel-jaw grippers. Its small form factor, high spatial resolution and open-source availability have established DIGIT as a de facto standard platform for tactile learning research and it is the sensor used, in simulated form, throughout this thesis.

The key property that motivates the use of DIGIT in this work is that its output is a standard RGB image. This means that the rich body of image-based representation learning literature, including Vision Transformers and contrastive pre-training, is directly applicable to tactile signals without any architectural modification. The TVL dataset [5], which provides the pre-trained tactile encoder used in this thesis, was collected entirely with DIGIT sensors, ensuring that the learned representations are directly compatible with the simulation stack described in the following section.

2.3 TACTO: Simulating Vision-Based Tactile Sensors

Training a cross-modal projection module requires large quantities of paired tactile-visual observations. Collecting these pairs on a physical robot is expensive, slow and difficult to reproduce consistently across different objects and contact configurations. **TACTO** [7] addresses this by providing a high-fidelity, real-time physics-based renderer that simulates

the optical formation process of image-based tactile sensors, including DIGIT, inside standard rigid-body physics engines.

The simulation works by intercepting the local contact mesh between the sensor's elastomeric surface and the target object at each timestep, then rendering the resulting deformation using a photometric model that replicates the sensor's internal LED illumination, camera optics and gel-elastomer response. The output is an RGB image that is visually and statistically close to what a real DIGIT sensor would produce under the same contact geometry, enabling sim-to-real transfer of learned tactile representations.

TACTO was originally released with native integration for PyBullet and has since been adopted across a range of physics backends in the tactile-manipulation literature. In this thesis, TACTO provides the tactile rendering layer on top of the physics simulation described in Chapter 6, allowing the data collection pipeline to run autonomously at scale and generate large numbers of synchronised tactile-visual pairs across a diverse set of object geometries without any physical hardware.

2.4 Cross-Modal Representation Alignment

Cross-modal representation alignment is the problem of learning a mapping between the embedding spaces of two different sensory modalities, such that semantically equivalent observations, regardless of their physical origin, are represented by nearby points in a common target space. In contrast to multimodal fusion, which combines several modalities that are simultaneously available as input, cross-modal alignment is concerned with making one modality's representation *stand in for* another's within a space that was originally defined by a single modality.

The most influential instance of this idea in the vision-language literature is CLIP [9], which aligns image and text representations through large-scale contrastive learning: an image encoder and a text encoder are trained jointly so that matching image-text pairs are pulled together in a shared embedding space while non-matching pairs are pushed apart. This contrastive formulation has since been extended beyond vision-language pairs; the TVL dataset and model [5] apply the same principle to tactile signals, training a Vision Transformer encoder to project tactile images directly into the CLIP latent space using paired touch-vision-language data, without requiring an intermediate adapter layer.

For this thesis, cross-modal alignment serves a specific and narrower purpose than in the general multimodal literature: rather than constructing a new shared embedding space from scratch, the goal is to project tactile representations directly into the *existing* visual latent space of a frozen VLA, so that the tactile-derived embedding can substitute for the visual one when vision is degraded or unavailable. This is a stricter requirement than generic cross-modal alignment, because the target space is fixed in advance and has been optimised exclusively for visual features during large-scale pretraining. Whether a lightweight projection module can satisfy this constraint, and how much paired data is needed to train it, are among the central research questions of the proposed work.

2.5 OpenVLA: Architecture and Visual Latent Space

OpenVLA [3] is a 7B-parameter open-source Vision-Language-Action model trained on 970k real-world robot demonstrations from the Open X-Embodiment dataset, spanning more than 20 different robot embodiments. It was selected as the target backbone for this thesis on the grounds of its fully open-source weights, well-documented architecture and active research community. Understanding its architecture is essential for designing a projection module whose output is compatible with the model's input expectations.

Visual encoder. OpenVLA uses a fused visual encoder consisting of a SigLIP and a DINOv2 backbone that maps image inputs to a sequence of image patch embeddings. The two encoders play complementary roles: SigLIP provides semantic alignment with language, while DINOv2 contributes spatial reasoning and fine-grained structural features. Their feature maps are concatenated channel-wise and projected through a two-layer MLP into the language model's embedding space, producing a rich multimodal representation that captures both high-level semantics and local spatial structure.

Language model and action head. A Llama 2 7B language model backbone predicts tokenised output actions, which are then decoded into continuous actions that can be directly executed on the robot. In the original OpenVLA formulation, actions are discretised into 256 bins per dimension and represented as special tokens appended to the standard vocabulary, allowing the model to treat action prediction as a standard autoregressive next-token prediction task. This discrete-token design is a key architectural constraint of the base model, with direct implications for inference speed and for the choice of learning objective: implications that motivate the OpenVLA-OFT variant described in Section 2.6.

Implications for the projection module. The tactile-to-visual projection head f_θ must produce embeddings that are compatible with the fused SigLIP–DINOv2 feature representation prior to the two-layer MLP projector. This defines the target dimensionality d_V and the distributional properties of $z_\mathcal{V}$: the projection head must match both the dimension and the statistical structure of the visual patch embeddings, so that the downstream MLP projector and language model backbone can process them without any architectural modification.

Frozen backbone as a design principle. In this thesis, the visual encoder and the language backbone are kept completely frozen throughout all training stages. This design choice is motivated not only by computational efficiency: it is a deliberate strategy that preserves the full generalisation capability acquired during large-scale pretraining on hundreds of thousands of robot demonstrations. Updating the backbone weights to accommodate tactile inputs would risk catastrophic forgetting of the visual-semantic priors that make the policy useful across tasks and environments. The entire adaptation budget is therefore allocated to the lightweight external modules, the tactile encoder and the projection head,

whose primary function is to bridge the gap between tactile and visual representations without altering the policy's core reasoning capability.

2.6 OpenVLA-OFT: Optimized Fine-Tuning

While the original OpenVLA formulation generates actions autoregressively as a sequence of discrete tokens, this design exhibits two practical limitations that are directly relevant to the present work: inference throughput is low, since actions must be decoded one dimension at a time and the discretisation of the action space into 256 bins can discard fine-grained control detail.

OpenVLA-OFT [11] addresses both limitations through an Optimized Fine-Tuning (OFT) recipe that modifies how the pretrained OpenVLA backbone is adapted, without altering its underlying pretraining. The recipe combines four design choices:

- **Parallel decoding.** The causal attention mask used during autoregressive generation is replaced with bidirectional attention over a set of empty action-query embeddings, allowing the entire action vector to be predicted in a single forward pass rather than one dimension at a time.
- **Action chunking.** By inserting additional empty action-query embeddings, the model predicts a short horizon of future actions in the same forward pass, amortising inference cost across multiple timesteps and providing smoother, temporally consistent behaviour.
- **Continuous action representation.** The discrete, bin-based action tokens of the original OpenVLA are replaced with a regression head that maps the final-layer hidden states directly to continuous action values, avoiding the quantisation artefacts introduced by discretisation.
- **L1 regression objective.** The model is fine-tuned with a simple L1 loss between predicted and ground-truth continuous actions, in place of the cross-entropy objective used for discrete token classification.

Together, these changes increase inference throughput by approximately 26 times relative to the original autoregressive OpenVLA, while simultaneously improving task success rate: the average success rate increased across the four LIBERO task suites (see Section 2.7) from 76.5% to 97.1%.

OpenVLA-OFT is adopted as the frozen backbone for the experimental setup developed in this thesis (Chapter 6). This choice is motivated by two considerations directly relevant to the proposed architecture. First, its continuous action representation and regression-based objective are more naturally compatible with a downstream policy consistency loss (Section 7.6) than the discrete token classification of the original OpenVLA, since gradients with respect to continuous action errors are well-defined without requiring differentiation through a discretisation step. Second, its substantially higher inference throughput makes iterative experimentation with the fusion mechanism described in Chapter 7 considerably

more practical within the time constraints of this project. As established in Section 2.5, the vision and language backbones remain frozen throughout; the OFT recipe affects only the action head and no backbone weight is altered during the tactile-projection training of Chapter 7.

2.7 The LIBERO Benchmark for Lifelong Robot Learning

LIBERO [8] is a simulation benchmark for studying knowledge transfer in multi-task and lifelong robot learning. It is built on top of the RoboSuite/MuJoCo physics stack and provides 130 language-conditioned manipulation tasks, together with high-quality human-teleoperated demonstrations for every task.

The benchmark is organised into four task suites, each designed to isolate a specific type of distribution shift:

- **LIBERO-Spatial:** tasks that share the same objects and goals but vary the spatial arrangement of the scene, testing transfer of spatial-relational knowledge.
- **LIBERO-Object:** tasks that share the same spatial layout and goal but introduce novel object instances, testing transfer of object-centric knowledge.
- **LIBERO-Goal:** tasks that share the same objects and spatial layout but require different goal states, testing transfer of task-goal knowledge.
- **LIBERO-100 (LIBERO-90 / LIBERO-Long):** a larger and more heterogeneous suite mixing all three types of knowledge, including ten long-horizon tasks that require executing several manipulation sub-skills in sequence.

LIBERO is selected as the evaluation environment for this thesis for three main reasons. First, it is the benchmark on which OpenVLA-OFT was originally fine-tuned and evaluated and officially released checkpoints are available for all four task suites, removing the need to collect a new demonstration dataset for baseline policy competence. Second, its procedurally generated task suites and controlled distribution shifts provide a natural setting for isolating the effect of visual degradation independently of other sources of task difficulty. Third, its manipulation tasks include pick-and-place and contact-rich sub-skills that are directly compatible with the integration of a simulated tactile sensor, as described in Chapter 6.

Because LIBERO does not natively provide tactile observations, the tactile modality used in this thesis is introduced by integrating a TACTO-simulated DIGIT sensor into the LIBERO simulation stack; the resulting environment configuration is described in detail in Chapter 6.

3

Related Work

3.1 Overview and Organisation of the Literature Review

This chapter situates the proposed approach within the broader research landscape spanning cross-modal visuo-tactile perception, cross-modal representation alignment and Vision-Language-Action models. The review is organised in two complementary parts.

Section 3.2 presents a historical excursus covering twenty-five works, tracing the evolution of the field from early cross-modal visuo-tactile perception, through the foundations of contrastive cross-modal alignment, the emergence of the VLA paradigm, tactile-language semantics and current benchmarking and robustness efforts, up to the most recent tactile-VLA research line. This part is intentionally broad rather than deep: each work is discussed briefly, in the context of the thematic cluster to which it belongs, to provide a comprehensive map of the research space.

Sections 3.3–3.7 then provide an in-depth analysis of five works selected for their direct methodological relevance to the research question formalised in Chapter 4. Each is examined in terms of its key contributions, architecture, strengths and weaknesses and, most importantly, the specific gap it leaves open that this thesis aims to address. Section 3.8 closes the chapter with a comparative summary mapping each of the five works to the specific components of the proposed architecture.

3.2 Historical Excursus

3.2.1 Early Cross-Modal Visuo-Tactile Perception

The idea that vision and touch can inform one another predates the VLA paradigm by nearly a decade. Falco et al. [12] were among the first to demonstrate cross-modal visuo-tactile object recognition through active robotic exploration, showing that a robot could improve its recognition accuracy by actively choosing where to touch based on visual cues. Around the same time, Yuan et al. [13] introduced a dataset directly associating visual and tactile properties of physical materials, establishing an early empirical link between surface appearance and tactile signature. Lee et al. [14] approached the problem from a generative angle, using conditional GANs to synthesise plausible tactile images from visual input and vice versa on the ViTac fabric dataset, while Li et al. [15] formalised cross-modal prediction between touch and vision more broadly, framing it explicitly as a translation problem between two sensory domains; a perspective that is conceptually related to the

projection task addressed in this thesis, albeit without any connection to downstream robot control. Calandra et al. [16] shifted the focus from perception to action, showing that combining vision and touch improves grasp success and regrasp decisions, an early demonstration that multimodal sensing has direct utility for manipulation, not only for recognition.

This period also saw the consolidation of the hardware and simulation infrastructure that later work, including this thesis, relies upon. Lambeta et al. [6] released DIGIT, establishing an open-source, low-cost optical tactile sensor that became a de facto standard. Wang et al. [7] subsequently released TACTO, enabling photorealistic simulation of such sensors without physical hardware. Closing this period, Guzey et al. [17] showed that tactile representations can be pre-trained in a self-supervised manner from unstructured robotic play, foreshadowing the representation learning strategies later adopted for tactile encoders in VLA-adjacent work, including the TVL encoder used in this thesis.

3.2.2 Foundations of Cross-Modal Representation Alignment

The methodological foundation for projecting one modality into the latent space of another comes from the vision-language literature. CLIP [9] demonstrated that a shared embedding space for images and text could be learned purely through contrastive pretraining on large-scale, weakly supervised data, without any explicit spatial or structural supervision. ImageBind [18] generalised this principle beyond a single pair of modalities, binding six sensory modalities to a shared image-centric embedding space and showing that contrastive alignment scales to modalities for which little or no paired data with every other modality exists. Fu et al. [5] (discussed in depth in Section 3.3) directly extended this contrastive paradigm to touch, training a tactile encoder to project into the existing CLIP latent space. This body of work establishes the core methodological premise of this thesis: that a fixed, pretrained embedding space can absorb a new modality through a lightweight learned projection, without retraining the space itself.

3.2.3 The Emergence and Evolution of VLA Models

The VLA paradigm began with RT-1 [10], which showed that a single transformer policy trained on a large and diverse corpus of real robot demonstrations could generalise substantially better than prior task-specific controllers. RT-2 [1] was the first to explicitly fuse this idea with pretrained vision-language models, representing actions as text tokens and fine-tuning a VLM directly on robot data, thereby transferring web-scale semantic knowledge to control. Subsequent work diversified the action-generation mechanism: π_0 [2] replaced autoregressive token prediction with a diffusion-based action expert, while OpenVLA [3] prioritised openness, releasing a 7B-parameter model and training recipe that substantially lowered the barrier to academic research and is adopted as the target backbone of this thesis. OpenVLA-OFT [11] later revisited the action decoding mechanism itself, replacing autoregressive discrete-token generation with parallel decoding, action chunking, and a continuous action representation, and is the specific OpenVLA variant used

throughout this work (Section 2.6). Ma et al. [19] provide a comprehensive survey of this rapidly evolving landscape, offering a taxonomy of VLA architectures, training strategies and evaluation practices that is useful for situating the narrower tactile-VLA line of work discussed in Section 3.2.6.

3.2.4 Tactile-Language Semantics

A parallel research direction has focused on connecting tactile sensing directly to natural language, rather than to a purely visual latent space. Yu et al. [20] introduced Octopi, a tactile-language model capable of reasoning about object properties, such as hardness and roughness, from a sequence of tactile frames and generating natural-language descriptions of contact. This line of work is relevant to this thesis because it demonstrates an alternative target representation for tactile signals (natural language descriptions rather than a continuous visual embedding), a design choice later revisited critically in Section 3.4, where linguistic tactile descriptions are shown to outperform raw tactile images when used as input to a high-level planner.

3.2.5 Benchmarks and Robustness in VLA Research

As VLA models matured, standardised benchmarks and robustness studies became necessary to compare architectures fairly and to expose their failure modes. Liu et al. [8] introduced LIBERO, a benchmark of 130 language-conditioned manipulation tasks organised into suites that isolate specific types of knowledge transfer (spatial, object, goal and long-horizon); LIBERO is adopted as the evaluation environment for this thesis (Section 2.7). Chi et al. [21] introduced Diffusion Policy, a visuomotor policy learning framework based on action diffusion that predates the VLA paradigm proper but remains a widely used non-VLA baseline in comparative evaluations, including several of the tactile-VLA works reviewed in Section 3.2.6. More recently, the LIBERO-Plus benchmark [22] extended the original LIBERO suite with an in-depth robustness analysis, systematically perturbing visual and task conditions to characterise how brittle current VLA policies are outside their training distribution: a concern directly relevant to the visual-degradation setting studied in this thesis.

3.2.6 Tactile-VLA Research Line

The most recent and most directly relevant cluster of work integrates tactile sensing into the VLA paradigm itself. Hao et al. [23] proposed VTLA, a Vision-Tactile-Language- Action model for high-precision insertion tasks, introducing a token ordering scheme that exploits recency bias in the underlying language model and applying preference-based optimisation to bridge the mismatch between classification-style training objectives and continuous control. Shah et al. [24] took an orthogonal approach with LITEN, which does not modify the VLA at all but instead learns, at inference time, the affordances and failure modes of a frozen VLA policy through structured causal assessment of past attempts: a purely

visual and linguistic feedback loop that does not involve touch, but which is conceptually relevant as a possible future robustness layer for uncertain tactile projections. More recently, OmniVTLA [25] proposed a vision-tactile-language-action model with semantically aligned tactile sensing and TacVLA [26] introduced contact-aware tactile fusion aimed at robustness under occlusion. Together with VLA-Touch and Tactile-VLA, discussed in depth below, this cluster represents the immediate research context of this thesis: a rapidly growing effort to combine tactile sensing with the VLA paradigm, within which the frozen-backbone, latent-substitution approach proposed here occupies a specific and, to date, largely unexplored position.

3.3 TVL: Touch-Vision-Language Alignment

Reference: Fu et al., ICML 2024 [5] **arXiv:** 2402.13232

Key Takeaways

- First large-scale dataset aligning tactile, visual and natural-language modalities: 43,741 in-contact pairs, collected with a DIGIT image-based tactile sensor and a co-mounted RGB camera.
- Pairwise contrastive learning across all three modality pairs (tactile-vision, tactile-language, vision-language) outperforms vision-centric binding (ImageBind) on tactile classification.
- A ViT tactile encoder trained to project directly into the CLIP latent space, without an intermediate adapter layer, achieves competitive performance with a modest fraction of human-annotated data, the remainder covered by pseudo-labels.
- The resulting encoder, integrated with a language model via LoRA, surpasses a strong vision-language baseline on the proposed tactile-semantic benchmark.

Architecture

- **Tactile encoder:** ViT (Tiny / Small / Base) initialised from scratch, trained to project tactile RGB images into the frozen CLIP latent space.
- **Training objective:** InfoNCE contrastive loss computed over all three modality pairs simultaneously.
- **Key regularisation:** background frames without contact are included during training to reduce overfitting to noisy pseudo-labels.

Positives and Negatives

+ Open-source dataset, code and pre-trained encoder weights: directly reusable as pre-training data and initialisation for this thesis.
+ Demonstrates feasibility of learning a tactile representation aligned to a fixed visual embedding space from a relatively modest amount of paired data.
– Static frames only, no temporal dynamics.
– Single sensor type (DIGIT); generalisation to other tactile hardware is not validated.
– No downstream robotics task: alignment quality is evaluated only on semantic classification and description generation, never on action generation.

Gaps and Relevance to This Thesis

- **Target space mismatch:** the encoder aligns tactile signals to the CLIP latent space, optimised for image-text retrieval, not for action generation. A frozen VLA uses its own vision encoder, whose representations carry task-relevant structure that CLIP does not preserve.
- **No action loop:** TVL never connects the aligned representation to a policy; the gap between tactile-semantic alignment and tactile-to-action transfer remains open.

3.4 VLA-Touch: Dual-Level Tactile Feedback

Reference: Bi et al., 2025 [27] **arXiv:** 2507.17294

Key Takeaways

- Modular dual-level framework that enhances a frozen VLA with tactile feedback at two complementary levels, without modifying any base model weights: high-level task planning via semantic tactile descriptions fed to a vision-language planner and low-level action refinement via a diffusion-based controller conditioned on tactile signals.
- Linguistic tactile descriptions outperform raw tactile images when provided to the vision-language planner, suggesting that the bottleneck for high-level reasoning is representation, not sensing itself.
- An interpolant-based controller refines VLA-generated action chunks by diffusing from the VLA's own output distribution rather than from Gaussian noise, improving task success over the base policy.
- Ablation confirms both levels are necessary: removing either the planning-level or the control-level tactile feedback causes a substantial drop in success rate.

Architecture

- **Base VLA:** frozen throughout, no fine-tuning with tactile data.
- **Tactile-language model:** processes a short tactile frame sequence and outputs continuous property scores or aggregated force vectors.
- **High-level planner:** a frozen vision-language model receives the scene image, task goal and tactile-derived text, producing primitive action instructions in a closed replanning loop.
- **Low-level controller:** an interpolant diffusion policy refines action chunks from the frozen VLA using visual features and the aggregated tactile vector.

Positives and Negatives

+ **Frozen VLA backbone:** the base policy is never modified, the closest architectural philosophy to this thesis among the works reviewed.
+ Systematic comparison of raw tactile images versus linguistic descriptions as planner input, directly informative for representation design.
– Tactile representation is weak: an aggregated force tuple discards spatial contact maps, texture and temporal dynamics beyond a few frames.
– Heavy external dependencies (a closed-source planner, a separate tactile-language model and the base VLA) create latency, cost and reproducibility concerns.
– Vision is always present; the system is never tested under degraded or absent visual input.

Gaps and Relevance to This Thesis

- **Tactile as augmentation, not substitution:** VLA-Touch adds tactile feedback on top of vision at both levels; it never considers the scenario where vision fails or is absent. This is the exact gap this thesis addresses by projecting tactile signals into the visual latent space, making the frozen VLA operable without a visual observation.
- **Handcrafted versus learned tactile representation:** the force-tuple encoding used for control is expert-designed and fixed. This thesis instead learns a projection $f_\theta : z_\mathcal{T} \to z_\mathcal{V}$ end-to-end.

3.5 Tactile-VLA: VLM Physical Knowledge for Tactile Generalization

Reference: Bi et al., 2025 [4] **arXiv:** 2507.09160

Key Takeaways

- Core insight: pretrained VLMs already contain latent semantic knowledge of physical interactions (e.g. what “gently” or “firmly” means in terms of force). A lightweight tactile interface trained on a comparatively small number of demonstrations is sufficient to activate this knowledge, without large-scale tactile pretraining.
- Force-language grounding is demonstrated zero-shot: a model trained on one insertion task with “soft” versus “hard” instructions transfers the corresponding force magnitudes to an unseen insertion task with a different object.
- A chain-of-thought variant reasons over tactile feedback, diagnoses failures and autonomously adjusts strategy, achieving strong zero-shot success on a task where all baselines fail.

Architecture

- **Backbone:** a generalist VLA policy; shared components (vision encoder, language tokenizer, transformer trunk) are initialised from pretrained weights.
- **Tactile encoder:** a simple MLP that compresses a history of force-vector measurements into a single fused token.
- **Fusion:** non-causal attention over a prefix sequence of vision, language and tactile tokens; all modalities cross- attend freely before reaching the action expert.
- **Action output:** an augmented action vector including an explicit force target, which is what forces the model to learn force-language grounding.

Positives and Negatives

+ Provides strong empirical evidence that VLMs contain latent physical knowledge activatable with minimal data, the key theoretical assumption underlying this thesis.
+ Data-efficient relative to other tactile-VLA approaches.
– Tactile encoder operates on scalar/vector force measurements, not on the rich RGB tactile images produced by DIGIT or similar sensors; a single token carries no spatial information.
– The backbone is partially fine-tuned (new modules added, action expert modified), not strictly frozen.
– All tasks are constrained; no contact-rich task requiring full spatial tactile awareness is evaluated.

Gaps and Relevance to This Thesis

- **Force vectors, not optical tactile images:** bridging the gap between scalar force measurements and rich RGB tactile images, by learning a projection from optical tactile images into the VLA latent space, is a central contribution of this thesis.
- **No visual latent substitution:** Tactile-VLA adds touch as an extra modality; vision is always present. The scenario where vision is degraded or absent, the primary motivation of this work, is never addressed.

3.6 Don't Blind Your VLA: Visual Representation Alignment

Reference: Kachaev et al., 2025 [28] **arXiv:** 2510.25616

Key Takeaways

- Systematic study showing that naive action fine-tuning of a pretrained VLM into a VLA degrades the original vision-language representations, even though the visual encoder weights are nominally preserved.
- Probing of hidden representations and attention maps reveals three concrete failure modes: attention sink (attention maps become diffuse and only weakly correlated with the target object), representation collapse (patch embeddings lose separability) and domain forgetting (vision-language understanding of symbolic or abstract categories degrades after supervised fine-tuning).
- The authors propose Visual Representation Alignment: during fine-tuning, the VLA's visual tokens are pulled toward the features of a frozen teacher vision-language model via a cosine-similarity loss applied through a lightweight, frozen projector, motivated by the Platonic Representation Hypothesis.
- This lightweight alignment loss mitigates the observed degradation and yields improved generalisation to out-of-distribution scenarios, without requiring any additional labelled data.

Architecture

- **Base VLA:** OpenVLA, used both as the object of study (to characterise representation degradation) and as the model to which the proposed alignment loss is applied.
- **Teacher representation:** features from a frozen, pretrained vision(-language) model, held fixed throughout training.
- **Alignment mechanism:** a lightweight, frozen projector maps the VLA's visual tokens toward the teacher's feature space; training minimises a cosine-similarity loss between the two, in addition to the standard action-prediction loss.

- **Evaluation:** representation probing (attention maps, embedding separability) combined with targeted tasks contrasting the VLA against its unmodified VLM counterpart and out-of-distribution manipulation evaluation.

Positives and Negatives

+ Directly diagnoses *why* a frozen-backbone or lightly-adapted VLA might still lose useful visual structure during adaptation: a risk this thesis must also account for, even though its own backbone is fully frozen rather than fine-tuned.

+ The alignment loss is lightweight (a single frozen projector and a cosine-similarity term), architecturally compatible with the kind of external, non-invasive modules this thesis relies on.

+ Rigorous probing methodology (attention maps, embedding separability) offers a template for diagnosing whether the tactile-derived projection in this thesis actually lands on the visual manifold, rather than merely minimising a training loss.

– The paper addresses representation degradation *within* the visual modality during fine-tuning; it does not consider an entirely different input modality (touch) or the scenario of degraded or missing visual input at inference time.

– The proposed alignment loss requires access to a separate teacher model at training time, introducing an additional dependency not required by the pretraining-only frozen backbone used in this thesis.

Gaps and Relevance to This Thesis

- **Different failure mode, same underlying concern:** this work studies degradation of visual representations induced by *fine-tuning*, whereas this thesis studies degradation of the visual *input signal itself*. Both, however, share the same underlying requirement: a representation fed into the frozen action-generation pathway must remain faithful to the structure the pathway was trained on.

- **Validated tool for projecting onto the appropriate manifold:** the cosine-similarity alignment mechanism proposed here is methodologically close to the cosine term used in the hybrid alignment loss of this thesis (Section 7.6), providing independent evidence that cosine-based alignment against a frozen target is an effective way to preserve semantic structure in a VLA's visual latent space.

3.7 Improving Robustness of VLA Models via Corrupted Visual Input Restoration

Reference: Guarnizo Orjuela et al., 2026 [29] **arXiv:** 2602.01158

Key Takeaways

- Quantifies a previously underexplored failure mode of VLA models: sensor-level image corruptions (electronic noise, dead pixels, lens contaminants), as distinct from physical occlusions caused by scene geometry.
- Demonstrates that state-of-the-art VLAs suffer catastrophic performance degradation under common signal artefacts, with success rates collapsing from approximately 90% to as low as 2% under some corruption types.
- Proposes the Corruption Restoration Transformer (CRT), a transformer-based module trained to restore a corrupted visual observation to its clean form *upstream* of the VLA, combining Shifted Patch Tokenization, Rotary Positional Embeddings and Locality Self-Attention with an adversarial training objective.
- Evaluates five distinct corruption types (including centred occlusion and Gaussian sensor noise) and shows that restoring the visual input with CRT largely neutralises their impact on downstream task success, without any modification to the VLA itself.

Architecture

- **Base VLAs evaluated:** frozen, off-the-shelf policies; the VLA is never modified or fine-tuned.
- **Restoration module (CRT):** a transformer-based image- to-image reconstruction model G that maps a corrupted observation x' to a restored estimate of the clean observation x, positioned entirely upstream of the VLA's own vision encoder.
- **Training objective:** adversarial training combined with a reconstruction objective, trained on paired clean/corrupted image data independent of the downstream VLA.
- **Inference pipeline:** the corrupted camera frame is first passed through CRT and only the restored image is fed to the unmodified, frozen VLA.

Positives and Negatives

+ Strictly non-invasive: the VLA backbone is never touched, mirroring the frozen-backbone philosophy adopted in this thesis and the restoration module is entirely modular and can in principle be paired with any VLA.
+ Provides a clear, quantified characterisation of a specific and realistic failure mode (sensor-level corruption) that is directly relevant to the visual degradation curriculum used in this thesis (Section 6).
+ Demonstrates that restoring the *visual* signal alone is sufficient to recover most of the lost performance for the corruption types studied, without requiring any other sensory modality.

- Restoration is only possible when the corrupted image still contains enough residual information to reconstruct the clean scene; it offers no solution when vision is *entirely* absent (e.g. full occlusion, sensor failure or a scenario with no camera at all), since there is nothing left to restore.
- The approach addresses sensor-level corruption specifically; it does not consider tactile or any other alternative modality as a source of information when restoration is not possible.
- Restoring an image is generally a harder and more data-intensive problem than substituting a compact latent embedding and provides no benefit in the fully occluded or blackout regime that this thesis specifically targets.

Gaps and Relevance to This Thesis

- **Complementary rather than competing failure regimes:** this work targets *corrupted-but-present* visual signals, recoverable at the pixel level; this thesis targets *degraded-to-absent* visual signals, where no amount of image restoration can recover the missing information and an alternative modality is required instead. The two approaches are naturally complementary and could, in principle, be combined along a single degradation spectrum: pixel-level restoration for mild corruption, tactile substitution for severe degradation or full blackout.
- **Precedent for a frozen, modular upstream fix:** CRT confirms that inserting an external, frozen-VLA-compatible module upstream of the visual encoder is a viable and effective strategy for improving robustness without touching backbone weights: directly supporting the architectural philosophy of this thesis, even though the module itself (image restoration versus cross-modal projection) differs.

3.8 Comparative Summary

Table 3.1 maps each of the five in-depth reviewed works to the specific components of the proposed thesis, clarifying how each contribution informs the design choices described in Chapter 7.

Table 3.1: Mapping of the five in-depth reviewed works to thesis components.

Paper	Key contribution to this thesis	Relevance to the proposed method
TVL		
Fu et al. [5]	*Foundation for tactile-visual-linguistic alignment.* Demonstrates that a tactile encoder can be trained to align with a fixed visual-linguistic latent space using paired touch-vision- language data.	The feature alignment loss $\mathcal{L}_{\text{feat}}$ proposed in this thesis builds directly on this principle. The TVL pre-trained encoder and dataset serve as a warm-start for the tactile branch.
VLA-Touch		
Bi et al. [27]	*Modular framework with frozen VLA backbone.* Demonstrates that tactile feedback can be added around a frozen VLA without modifying its weights, preserving generalisation.	The closest architectural precedent to the proposed approach. The frozen-backbone principle is adopted directly; this thesis goes further by projecting tactile signals into the VLA's visual latent space, enabling operation without any visual observation.
Tactile-VLA		
Bi et al. [4]	*Evidence of latent physical knowledge in VLMs.* Shows that VLMs already encode semantic understanding of physical interactions, activatable with comparatively little data.	Core empirical support for the thesis hypothesis: a frozen VLA that already "understands" force concepts should be able to generate coherent actions from a well-aligned tactile representation, without retraining its semantic layers.
Don't Blind Your VLA		
Kachaev et al. [28]	*Diagnosis and repair of visual representation degradation.* Shows that naive adaptation degrades a VLA's visual representations and proposes a lightweight cosine-alignment loss against a frozen teacher to mitigate this.	Provides both a diagnostic methodology (representation probing) and independent validation of cosine-based alignment against a frozen target, directly informing the alignment loss and evaluation protocol used for the tactile-to-visual projection.
Corrupted-Input Restoration		
Guarnizo Orjuela et al. [29]	*Quantifies and repairs sensor-level visual corruption upstream of a frozen VLA.* Shows that restoring corrupted images before they reach the VLA recovers most of the lost task performance, without touching the backbone.	Validates the frozen, modular, upstream-fix philosophy of this thesis, while leaving open exactly the regime this thesis targets: severe degradation or full visual blackout, where no amount of pixel-level restoration can recover the missing information.

4

Thesis Objective

4.1 Research Question

This thesis addresses the following research question:

> *Can a frozen Vision-Language-Action model maintain task performance under degraded or absent visual input by dynamically fusing tactile-derived representations into its visual latent space, without any modification to the base policy weights?*

The question is deliberately constrained along three dimensions that recur throughout the remainder of this thesis. First, *frozen*: the base policy weights (vision encoder, language backbone and action head) are never updated; all adaptation is confined to lightweight external modules. Second, *degraded or absent*: the object of study is not multimodal fusion under normal operating conditions, but the specific regime in which the visual channel becomes partially or fully uninformative. Third, *visual latent space*: the mechanism under investigation is not the addition of a new input stream alongside vision, but the substitution of the visual embedding with a tactile-derived counterpart that occupies the same representational role within the frozen policy.

4.2 Hypothesis

The central hypothesis of this thesis is that the visual latent space of a pretrained VLA is sufficiently general to accommodate representations derived from tactile sensing and that a lightweight learned projection can produce tactile embeddings that are behaviourally equivalent to visual ones when vision is unreliable. Under nominal visual conditions, the system is expected to reduce to the original, unmodified VLA behaviour; as vision deteriorates, the tactile-derived representation is expected to progressively take over the role normally played by the visual embedding, preserving task performance rather than causing an abrupt failure.

This hypothesis is grounded in two empirical observations drawn from the literature reviewed in Chapter 3. First, Fu et al. [5] demonstrate that a ViT encoder can be trained to align tactile images with a fixed visual-linguistic embedding space through contrastive learning on a comparatively modest amount of paired data, suggesting that the target space does not need to be redesigned to accommodate a new modality. Second, Bi et al. [4] show that frozen vision-language backbones already encode latent knowledge of

physical interactions, activatable with minimal tactile-specific supervision, suggesting that the downstream reasoning and action-generation machinery of a frozen VLA does not need to be retrained to make productive use of a tactile-derived input. Taken together, these results suggest that the semantic structure of a pretrained VLA's visual space is compatible with tactile-derived representations and that the policy's action-generation capability can plausibly be preserved without retraining, provided the tactile-to-visual projection is sufficiently accurate.

It is important to state explicitly what this hypothesis does *not* claim. It does not claim that a tactile representation can replace vision under all conditions or for all tasks: tactile sensing is inherently local and contact-dependent and cannot substitute for the global scene understanding that vision provides during, for instance, the approach phase of a manipulation task. The hypothesis is instead restricted to the specific and practically important regime in which vision is degraded or absent *during phases of the task where contact is already established or imminent*, precisely the phases in which tactile information is most informative and vision is least reliable.

4.3 Specific Goals

The research question is decomposed into three specific, empirically tractable goals.

1. **Feasibility of cross-modal alignment.** Before committing to a full training and evaluation pipeline, it must first be established whether tactile and visual embeddings occupy a compatible region of representational space and whether the gap between them is of a magnitude that a lightweight projection module can plausibly close. This goal is addressed through the preliminary exploratory analysis presented in Chapter 5.

2. **Alignment quality.** A tactile-to-visual projection module $f_\theta : z_\mathcal{T} \to z_\mathcal{V}$ should be trainable such that the projected tactile embedding is geometrically close to the visual embedding produced by the frozen VLA's vision encoder on the corresponding RGB observation. Alignment quality is to be measured by cosine similarity and mean squared error in the latent space, on a held-out set of paired tactile-visual observations.

3. **Infrastructure for policy-level evaluation.** A reproducible simulation environment must be assembled in which the substitution of $f_\theta(z_\mathcal{T})$ for the visual embedding $E_{\text{vision}}(I)$ can, in future work, be evaluated for its effect on action generation and task success under progressively degraded visual conditions. This goal is addressed by the benchmark setup described in Chapter 6, which integrates the LIBERO benchmark, a frozen OpenVLA-OFT backbone and a TACTO-simulated DIGIT sensor.

4.4 Scope and Design Choices

The target VLA is **OpenVLA-OFT** [11], selected for its fully open-source weights, its continuous action representation and its substantially higher inference throughput relative to the

original OpenVLA, all of which make it a practical frozen backbone for iterative experimentation (Section 2.6). The tactile sensor is **DIGIT** [6], simulated via the TACTO photometric renderer [7]. The evaluation environment is **LIBERO** [8], selected because officially released OpenVLA-OFT checkpoints already exist for its task suites, removing the need to collect a new baseline demonstration dataset and because its procedurally controlled task suites provide a setting in which visual degradation can be studied independently of other sources of task difficulty. Both the pretrained tactile encoder and the target visual encoder are kept entirely frozen throughout this work; the only trainable component under consideration is the projection module f_θ and its associated substitution mechanism.

5

Preliminary Studies on Tactile and Visual Data

Before committing to the full training of the projection module f_θ (Chapter 7), it is necessary to establish whether the raw ingredients of the alignment problem are, in fact, compatible: does the frozen visual encoder produce a target space with enough structure to be worth aligning to and does a candidate tactile encoder produce embeddings that share enough of that structure for a lightweight projection to close the gap between them? This chapter reports a small-scale exploratory study designed to answer these questions empirically, ahead of any commitment to a specific tactile backbone or a full-scale training run. All experiments in this chapter use a set of $N = 80$ paired tactile-visual samples (8 objects $\times$ 10 grasp poses each), collected with the pipeline described in Section 5.1.

5.1 Exploratory Setup and Object Design

The exploratory dataset was collected in the same simulated environment used throughout this thesis: a Franka Emika Panda gripper instrumented with two simulated DIGIT sensors [6] (one per finger), rendered via TACTO [7], alongside a third, uninstrumented external camera providing the RGB view consumed by OpenVLA-OFT's frozen vision backbone. For each of eight objects, the gripper was closed around the object from ten randomly perturbed grasp poses, yielding ten samples per object and 80 samples in total. For every sample, three images were saved: the left-finger DIGIT frame, the right-finger DIGIT frame and the external camera frame (Figure 5.1), together with proprioceptive metadata (end-effector pose, joint configuration and per-finger contact force).

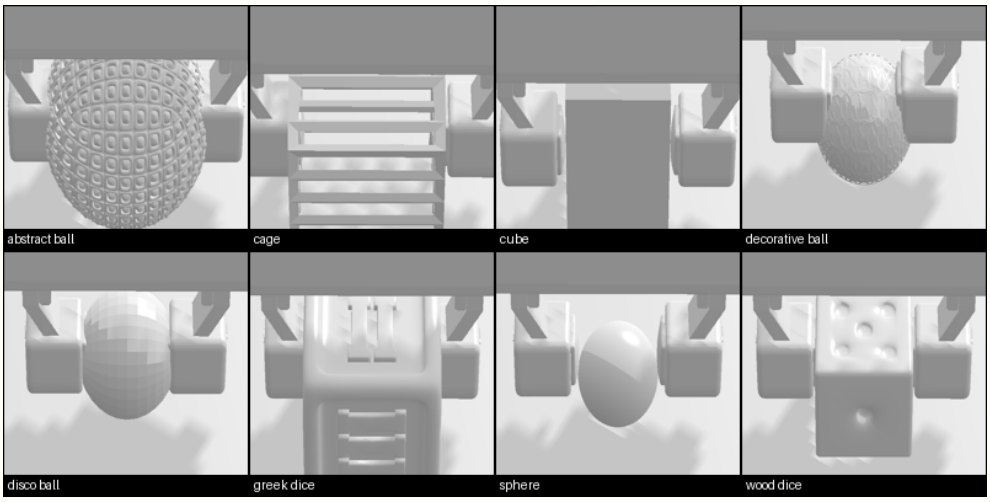


Figure 5.1: The eight objects used in the exploratory study, as seen from the external camera. Object selection deliberately varies two independent axes: overall geometry (solid convex shapes, faceted polyhedra and an open lattice) and surface texture (smooth, mirrored, engraved, grained or ornamentally patterned).

The eight objects were not chosen arbitrarily. They were designed to separate two properties that jointly determine what a tactile sensor can and cannot distinguish: coarse shape and fine surface detail. `sphere` and `cube` act as geometric baselines with essentially flat, uniform surfaces and no fine texture, isolating the contribution of gross shape alone. `disco_ball` preserves the spherical envelope of `sphere` but breaks its surface into small mirrored facets, testing sensitivity to fine faceting at constant global geometry. `decorative_ball` and `abstract_ball` both carry continuous, smoothly varying ornamental patterns of differing spatial frequency, again on a spherical envelope. `wood_dice` and `greek_dice` share a cuboid envelope similar to `cube` but add, respectively, a grained material texture with recessed pips and an engraved geometric relief, testing whether locally-planar geometry with superimposed fine detail is distinguishable from both the plain cube and from the patterned spheres. Finally, `cage` departs from all of the above with an open lattice structure that presents no continuous contact surface at all, acting as a geometric outlier against which the other seven, topologically closed objects can be contrasted. This design was intended to expose, already at this preliminary stage, whether a candidate tactile encoder responds primarily to gross geometry, to fine surface detail, or, as Section 5.2 shows to be at least partially true, fails to distinguish between the two at all.

Figure 5.2 shows representative left/right DIGIT frames for four contrasting objects, illustrating the range of contact patterns present in the dataset: from the near-uniform elastomer deformation of `sphere`, through the faceted highlights of `disco_ball` and the sparse, discontinuous contact points of `cage`, to the locally planar imprint of `wood_dice`.

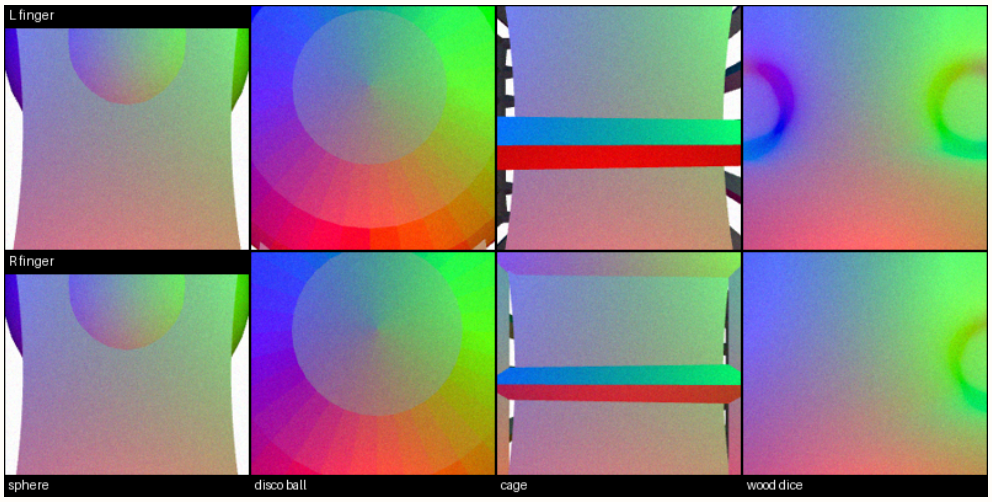


Figure 5.2: Representative left- and right-finger DIGIT frames for four contrasting objects, spanning uniform, faceted, sparse and planar contact patterns.

5.2 Tactile Signal Characterisation

The first question is whether a candidate tactile encoder even produces a usable signal: embeddings that are numerically well-behaved and that group consistently by object identity. Two candidate frozen tactile encoders were considered: (i) the official pre-trained tactile encoder released with the Touch-Vision-Language (TVL) framework [5], trained natively on real DIGIT data with a cross-modal contrastive objective and (ii) a generic ViT-Small Vision Transformer backbone, pre-trained on ImageNet only, with no exposure to tactile data of any kind. For both encoders, the two per-sample DIGIT frames were processed independently and the resulting embeddings averaged, producing a single tactile descriptor $z_{\mathcal{T}} \in \mathbb{R}^{384}$ per sample.

Signal quality was first assessed structurally, per object, via the mean pairwise cosine similarity within each object class (i.e. the diagonal of the inter-class similarity matrix introduced in Section 5.4). For the generic ViT-Small encoder, intra-class cosine similarity ranged from 0.9701 (`cage`, the geometric outlier with the most variable contact pattern) to 0.9980 (`abstract_ball`), indicating that repeated grasps of the same object are mapped to consistent, tightly clustered embeddings while still leaving room for inter-object separation. The same computation for the official TVL encoder yields comparably high, in places even higher, intra-class consistency (up to 0.9999 for `cage`). Taken in isolation, intra-class consistency would suggest that both encoders are equally viable candidates. As Section 5.4 shows, however, this metric alone is misleading: it cannot distinguish an encoder that is consistent *because* it correctly identifies object-specific contact geometry from one that is consistent because it maps essentially everything to the same region of the embedding space, regardless of what is actually being touched.

5.3 Visual Embedding Distribution Analysis

The visual target space is produced by feeding each sample's external camera frame through OpenVLA-OFT's frozen, fused SigLIP–DINOv2 vision backbone (Section 2.5), spatially average-pooling the resulting 256 patch tokens to obtain a single embedding $z_{\mathcal{V}} \in \mathbb{R}^{2176}$ per sample. Across all 80 samples, $z_{\mathcal{V}}$ is numerically well-behaved: the embedding norm has mean 95.87 and standard deviation 1.86 (range $[91.63, 99.09]$), individual coordinate values have mean 0.098 and standard deviation 2.05 (range $[-26.52, 29.98]$). No NaN or infinite values were observed in any of the 80×2176 entries. This confirms that the frozen vision backbone behaves as expected on the rendered scenes used in this study and that $z_{\mathcal{V}}$ is a stable regression target for f_θ.

More importantly, $z_{\mathcal{V}}$ exhibits a non-trivial per-object structure rather than collapsing to a single scene-level descriptor. Intra-class cosine similarity (repeated grasps of the same object) is very high, ranging from 0.9901 (`cage`) to 0.9993 (`abstract_ball`). It was expected, since the external camera view is dominated by the shared scene background, gripper and lighting, with only a small central region of the frame varying with the object in hand. Critically, however, the *inter*-class similarity is markedly lower and highly non-uniform across object pairs, with a global mean of 0.7634 (Section 5.4, Figure 5.3). Geometrically and texturally similar objects are placed closer together in $z_{\mathcal{V}}$ than dissimilar ones: the two smooth spheres `disco_ball` and `sphere` share 0.9455 cosine similarity, while the open lattice `cage` is comparatively distant from every other object (mean similarity ≤ 0.70 to all seven closed shapes). This confirms that, despite the small in-frame footprint of the manipulated object, the frozen visual encoder retains meaningful, object-specific structure: exactly the kind of structure that a tactile encoder would need to reproduce for cross-modal alignment to be feasible.

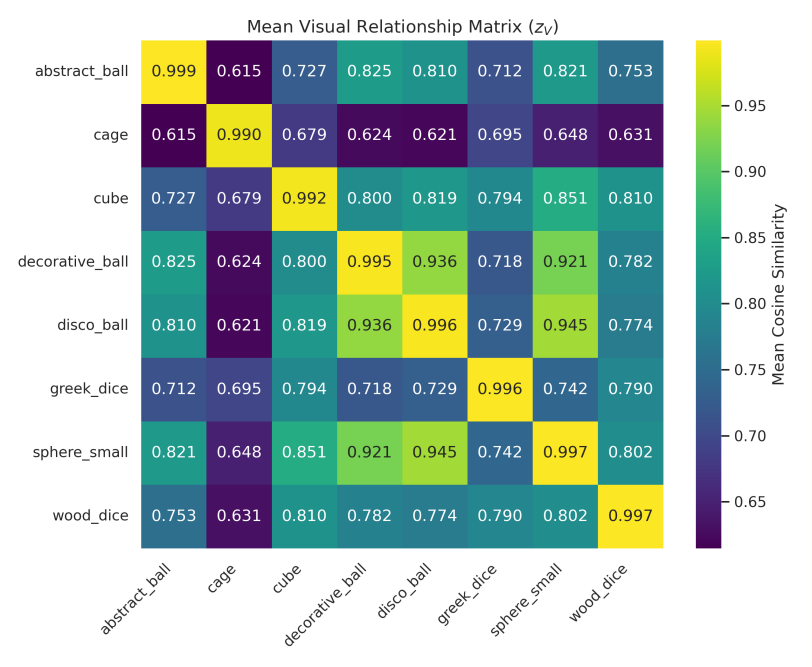


Figure 5.3: Inter-class cosine similarity heatmap for the visual embedding space $z_{\mathcal{V}}$ across the eight objects. Darker cells indicate lower similarity; the open-lattice `cage` object stands out as the most dissimilar from the other seven, closed-surface objects.

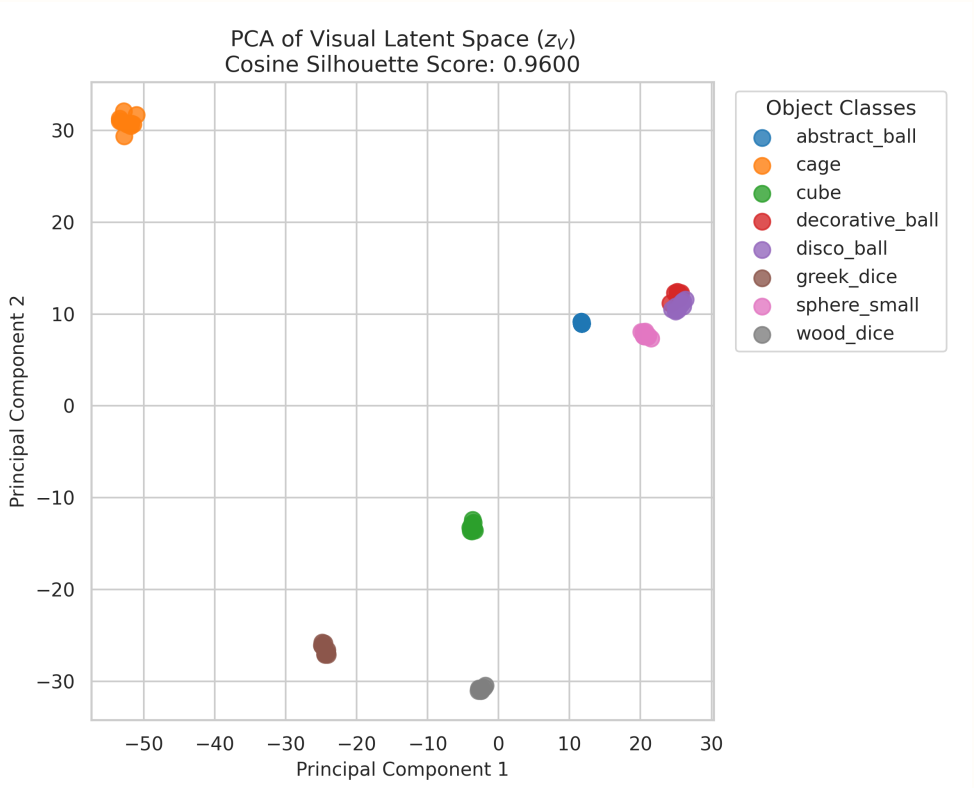


Figure 5.4: First two principal components of the visual embedding space $z_{\mathcal{V}}$ across the eight objects. Clusters are tight and well separated, consistent with the high intra-class and moderate inter-class cosine similarity reported in the text.

5.4 Modality Gap Quantification

Characterising each modality in isolation (Sections 5.2 and 5.3) is not sufficient to establish feasibility: what matters for f_θ is whether the *relational* structure of the tactile space, which objects are similar to which, agrees with the relational structure of the visual space. This was quantified using a representational similarity analysis (RSA), which compares relational structure rather than absolute feature values: for each modality, an 8×8 inter-class cosine similarity matrix was computed by averaging pairwise cosine similarities between all samples of each pair of object classes (Tables 5.1 and 5.2) and the two matrices were then compared directly.

Table 5.1: Inter-class cosine similarity matrix for the visual embedding space $z_\mathcal{V}$. Diagonal entries are intra-class similarities.

	abs. ball	cage	cube	dec. ball	disco	gr. dice	sphere	wood dice
abstract_ball	0.999	0.615	0.727	0.825	0.810	0.712	0.821	0.753
cage	0.615	0.990	0.679	0.624	0.621	0.695	0.648	0.631
cube	0.727	0.679	0.992	0.800	0.819	0.794	0.851	0.810
decorative_ball	0.825	0.624	0.800	0.995	0.936	0.719	0.921	0.782
disco_ball	0.810	0.621	0.819	0.936	0.996	0.729	0.946	0.774
greek_dice	0.712	0.695	0.794	0.719	0.729	0.996	0.742	0.790
sphere	0.821	0.648	0.851	0.921	0.946	0.742	0.997	0.802
wood_dice	0.753	0.631	0.810	0.782	0.774	0.790	0.802	0.997

Table 5.2: Inter-class cosine similarity matrix for the tactile embedding space $z_{\mathcal{T}}$, generic ViT-Small encoder. Diagonal entries are intra-class similarities.

	abs. ball	cage	cube	dec. ball	disco	gr. dice	sphere	wood dice
abstract_ball	0.998	0.384	0.358	0.300	0.329	0.349	0.276	0.403
cage	0.384	0.970	0.569	0.364	0.571	0.727	0.468	0.435
cube	0.358	0.569	0.993	0.460	0.694	0.656	0.587	0.732
decorative_ball	0.300	0.364	0.460	0.997	0.545	0.383	0.563	0.438
disco_ball	0.329	0.571	0.694	0.545	0.998	0.516	0.624	0.483
greek_dice	0.349	0.727	0.656	0.383	0.516	0.995	0.473	0.542
sphere	0.276	0.468	0.587	0.563	0.624	0.473	0.995	0.396
wood_dice	0.403	0.435	0.732	0.438	0.483	0.542	0.396	0.997

Two global summary statistics were derived from these matrices: the cosine silhouette score, measuring how well each modality separates the eight objects internally and the mean off-diagonal (inter-class) cosine similarity, measuring how tightly the whole embedding population is packed regardless of class. These are reported for all three spaces: visual, tactile (ViT-Small) and tactile (official TVL) in Table 5.3.

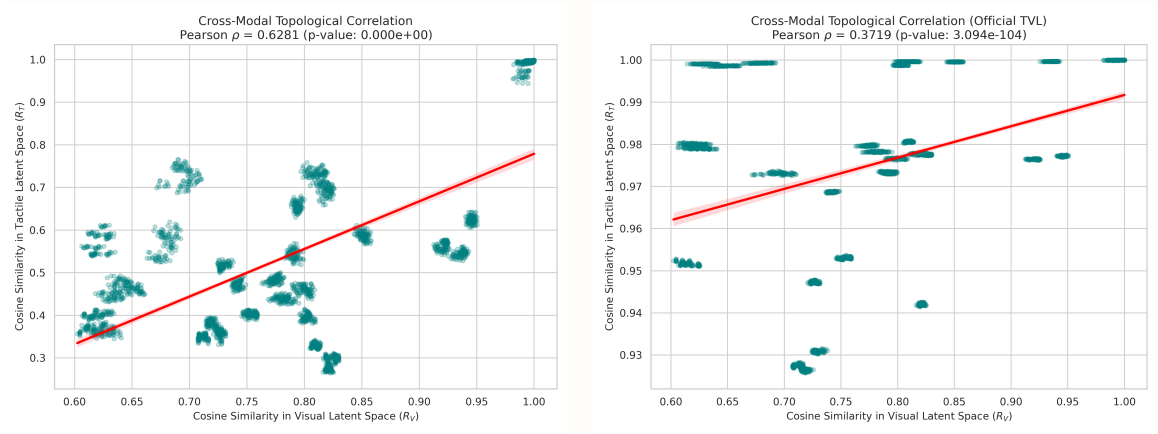
Table 5.3: Global separability metrics for the visual space and the two candidate tactile spaces.

Metric	$z_{\mathcal{V}}$ (target)	$z_{\mathcal{T}}$ (ViT-Small)	$z_{\mathcal{T}}$ (official TVL)
Cosine silhouette score	0.9600	0.9749	0.9769
Mean inter-class cosine similarity	0.7634	0.4867	0.9730
RSA correlation with $z_{\mathcal{V}}$ (Pearson ρ)	—	0.628	0.372

The silhouette scores alone would suggest that all three spaces separate the eight objects comparably well and would not favour one tactile encoder over the other. The mean inter-class similarity and, especially, the RSA correlation tell a different and more decisive story. The official TVL encoder packs all eight objects into an extremely narrow cone of the embedding hypersphere: mean inter-class similarity is 0.9730 and even the least similar pair in the dataset exceeds 0.92. The compression is so severe that geometrically very different objects such as `cage` and `cube` still reach 0.9993 similarity. The generic ViT-Small encoder, in contrast, spreads the same eight objects across a much wider range, with a mean inter-class similarity of only 0.4867 and individual pairs ranging from 0.276 (`abstract_ball` vs. `sphere`) to 0.732 (`cube` vs. `wood_dice`: the two objects sharing the cuboid envelope, as

intended by the object design of Section 5.1).

This difference is made directly visible by correlating the two inter-class similarity matrices against the visual one, sample-pair by sample-pair (Figure 5.5). For the ViT-Small encoder, the relational geometry of the tactile space tracks that of the visual space reasonably well (Pearson $\rho = 0.628$, $p < 10^{-3}$): pairs of objects that are visually similar tend also to be tactilely similar and pairs that are visually dissimilar tend to be tactilely dissimilar as well. For the official TVL encoder, this relationship is far weaker ($\rho = 0.372$) and visibly degenerate: nearly all tactile similarity values are compressed into the $[0.92, 1.00]$ band regardless of the corresponding visual similarity, which ranges over $[0.60, 1.00]$.



(a) Generic ViT-Small encoder (b) Official TVL encoder

Figure 5.5: Representational similarity analysis: cosine similarity in the tactile space (R_T) plotted against cosine similarity in the visual space (R_V) for every object pair. The generic ViT-Small encoder (a) preserves a visible positive relationship between the two modalities; the official TVL encoder (b) compresses almost all tactile similarities into a narrow band near 1.0, largely independent of the corresponding visual similarity.

This pattern is consistent with the phenomenon sometimes described in the representation-learning literature as anisotropic collapse: an encoder whose output occupies only a narrow cone of its embedding space, such that most pairwise similarities are uniformly high irrespective of semantic content. A plausible alternative explanation, however, cannot be ruled out with the present data. The official TVL encoder was trained exclusively on real DIGIT captures, whereas the tactile frames used in this study are produced by the TACTO photometric simulator; a sim-to-real domain gap could equally well explain why the geometric variance that TVL is known to encode on real hardware fails to activate on simulated input, without implying that the representation itself is degenerate in any general sense. Distinguishing between these two explanations, intrinsic anisotropy versus a simulation-induced domain gap, is not possible from this dataset alone and is left as an open question for future work, ideally to be revisited once real DIGIT data becomes available. For the purposes of the present thesis, however, this distinction is immaterial: irrespective of its cause, the official TVL encoder does not currently provide a tactile signal from which f_θ could learn a well-conditioned mapping onto $z_\mathcal{V}$ and is therefore not a viable choice under

the present experimental conditions.

Figure 5.6 visualises both modalities jointly via a shared t-SNE projection (computed on the ViT-Small tactile embeddings and the visual embeddings after independent PCA whitening to 50 components each). Same-object markers of the two modalities do not overlap: confirming that a modality gap is indeed present, exactly as expected, but they are arranged in a broadly consistent relative layout, with visually and tactilely similar objects (e.g. the three smooth-surfaced spheres) remaining neighbours across modalities. This is the qualitative signature of a gap that is well-suited to correction by a learned projection, as opposed to a gap masking two entirely unrelated geometries.

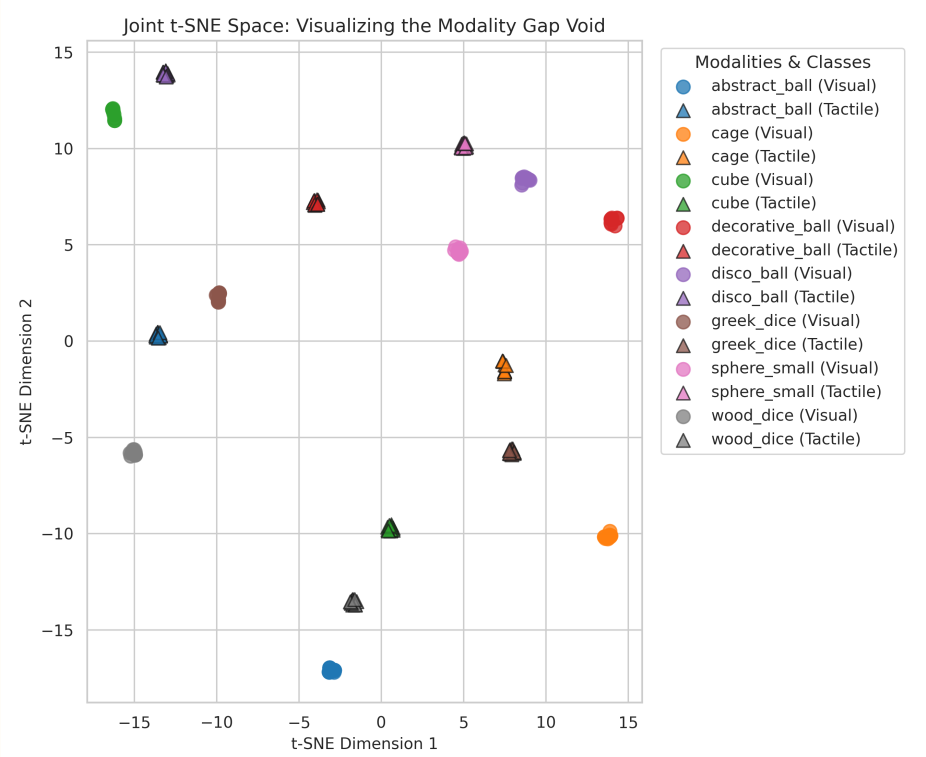


Figure 5.6: Joint t-SNE projection of the visual embeddings (circles) and ViT-Small tactile embeddings (triangles), colour-coded by object. The two modalities occupy distinct regions of the projection, the modality gap f_θ is meant to close, while preserving a broadly consistent relative arrangement of object classes across modalities.

5.5 Feasibility of Cross-Modal Alignment

The results of this chapter support, with two important qualifications, a positive answer to Goal 1 of Chapter 4: that tactile and visual embeddings occupy a compatible enough region of representational space for a lightweight projection module to plausibly close the gap between them. Three pieces of evidence support this conclusion. First, neither modality collapses numerically: $z_\mathcal{V}$ is stable and free of degenerate values (Section 5.3) and a generic ViT-Small tactile encoder produces embeddings with comparable numerical well-behavedness. Second, the visual target space retains substantial object-specific

structure despite the small in-frame footprint of the manipulated object, with inter-class similarity varying meaningfully across object pairs (Section 5.3) rather than collapsing to a single scene descriptor. Third, and most importantly, the relational geometry of the tactile space produced by a generic ViT-Small encoder correlates positively and non-trivially with that of the visual space (RSA $\rho = 0.628$), indicating that the two modalities already agree, to a first approximation and prior to any training of f_θ, on which objects are similar to which.

This conclusion, however, is conditional on the choice of tactile backbone, not an intrinsic property of tactile sensing in general. The natural first candidate, the official pre-trained TVL encoder, was found to exhibit a near-degenerate inter-class structure on the present simulated dataset (mean inter-class similarity 0.973, RSA $\rho = 0.372$), which would force f_θ into a severely ill-conditioned regression problem: mapping near-identical inputs onto well-separated visual targets. On the evidence gathered here, this thesis therefore adopts the generic ViT-Small encoder, rather than the official TVL encoder, as the frozen tactile backbone for the projection architecture described in Chapter 7. This choice is explicitly data-driven and provisional: it reflects behaviour observed on TACTO-simulated DIGIT frames and should be re-evaluated should the experimental pipeline be extended to real tactile hardware, where the domain gap discussed in Section 5.4 would no longer apply.

Two limitations of this preliminary study should be kept in mind when interpreting these results. First, the sample size–80 observations across eight objects–is sufficient to characterise gross distributional and relational properties of the two embedding spaces, but is far too small to train or validate f_θ itself; it serves here purely as a feasibility probe. Second, all conclusions are necessarily backbone and simulator specific: they establish that *a* viable tactile encoder exists for *this* simulated setup, not that any tactile encoder would work, nor that the same conclusion would transfer unchanged to real hardware. With these caveats stated, the outcome of this chapter is sufficient to justify proceeding, in Chapter 7, to the design and full-scale training of the projection module f_θ on top of the generic ViT-Small tactile backbone.

6

Benchmark Simulation Setup

While Chapter 5 established, on a small exploratory dataset, that a tactile and a visual embedding space are compatible enough for a projection module f_θ to plausibly bridge them, that analysis was carried out entirely outside the benchmark in which f_θ would ultimately need to operate. This chapter describes the construction and validation of that benchmark: the physics-simulated evaluation environment, the frozen VLA backbone under test, the custom Panda gripper instrumented with simulated DIGIT sensors and the software stack tying them together. Unlike Chapter 5, which reported embedding-level statistics, this chapter reports *policy-level* evidence (full episodic rollouts of a real manipulation benchmark) and is the first point in the thesis where the physical consequences of the gripper modification required for tactile sensing become measurable.

6.1 LIBERO as the Evaluation Environment

LIBERO [8] was adopted as the evaluation environment for this thesis, specifically the `libero_object` task suite: ten language-conditioned pick-and-place tasks sharing a fixed scene layout (objects arranged on a tabletop, to be picked up and placed in a basket) but differing in the target object (*alphabet soup*, *cream cheese*, *salad dressing*, *bbq sauce*, *ketchup*, *tomato sauce*, *butter*, *milk*, *chocolate pudding* and *orange juice*). This suite was chosen over the other LIBERO suites (`spatial`, `goal`, `10`, `90`) because it isolates object identity as the main source of task variation while keeping scene layout and required skill (grasp-and-place) constant; precisely the axis of variation this thesis is concerned with, since tactile sensing is expected to matter most when the geometry of the object in hand changes.

The environment is instantiated through LIBERO's own `OffScreenRenderEnv` wrapper, which in turn builds on robosuite and MuJoCo for physics simulation and rendering. Each of the ten tasks was evaluated over 50 independent trials (fixed initial-state distribution, one language instruction per task), for a total of 500 episodes per full evaluation run; the same protocol used for both results reported in this chapter. A successful episode is one in which the target object is placed in the basket within a task-specific step budget (280 steps for `libero_object`, following the convention established in the OpenVLA-OFT evaluation protocol).

6.2 OpenVLA-OFT as Frozen Backbone

The VLA policy under test is OpenVLA-OFT [11] (Section 2.5), evaluated using the official pre-trained checkpoint fine-tuned specifically on `libero_object` (`moojink/openvla-7b-oft-finetuned-libero-object`), loaded through the official `openvla-oft` repository at `github.com/moojink/openvla-oft`, following its documented LIBERO evaluation protocol. Throughout every experiment in this thesis, the entire 7-billion-parameter model – vision backbone, projection layers, and language-conditioned action head alike – remains completely frozen; it is used exclusively as a fixed policy under test and as the source of the target visual embedding $z_{\mathcal{V}}$ introduced in Section 5.3.

Before any modification to the robot's gripper, a baseline evaluation was run to establish a reference success rate against which all subsequent changes could be measured. Over 500 episodes (10 tasks $\times$ 50 trials), the frozen policy achieved an overall success rate of **98.2**% ($491/500$), with per-task success ranging from 92% (*chocolate pudding*) to 100% (five tasks), consistent with the success rates reported in the original OpenVLA-OFT publication. This run used the stock robosuite Panda gripper, with no tactile sensors attached and establishes the ceiling performance against which the gripper substitution described in Section 6.4 is evaluated.

6.3 TACTO-MuJoCo: Tactile Sensor Simulation

Simulated tactile sensing is provided by TACTO [7], which renders optically plausible DIGIT [6] sensor frames from the local contact geometry between an elastomer gel surface and the object being touched. The exploratory dataset of Chapter 5 used TACTO in its native, standalone PyBullet-based rendering pipeline. The benchmark environment described in this chapter, however, runs on MuJoCo/robosuite (Section 6.1), a different physics and rendering stack; TACTO's PyBullet-specific rendering pipeline is not natively compatible with a MuJoCo scene graph and required a separate integration layer, based on the community TACTO–MuJoCo bridge (`L3S/TACTO-MuJoCo`), to expose the same gel-deformation and image-rendering pipeline inside a robosuite environment.

This integration was completed and validated directly inside the benchmark environment described in this chapter: paired `digit_left` and `digit_right` frames were confirmed to render correctly and to respond to contact during real grasp trials, rather than only in the isolated, static scenes used for the initial gripper smoke tests (Section 6.4). The results reported in Section 6.4 were collected with this tactile rendering pipeline active throughout every episode, alongside the standard `agentview` and wrist-camera observations consumed by the frozen OpenVLA-OFT policy.

6.4 Robot Platform and Scene Configuration

The robot platform is a Franka Emika Panda arm (7 DoF) with a parallel-jaw gripper, matching the platform assumed by the OpenVLA-OFT checkpoint used in this thesis. To accommodate tactile sensing, the stock gripper fingers were augmented with two simulated DIGIT sensors, one per fingertip, rigidly attached via fixed joints so that no additional degrees of freedom and no change to the existing position-controlled gripper actuation, are introduced. Each sensor is offset roughly 3.5 cm forward along the finger and rotated so that the two gel surfaces face one another, i.e. directly in the path of any object grasped between the fingers: the same design used for the exploratory data collection of Chapter 5, here ported from a standalone URDF/PyBullet description to a MJCF gripper model integrated into robosuite's gripper registry, so that it can be selected as a drop-in replacement for the stock Panda gripper without modifying robosuite itself or any other part of the OpenVLA-OFT evaluation pipeline. The external `agentview` camera and wrist camera used by the policy were left untouched by this substitution.

Before running the full evaluation, the modified gripper was validated in isolation (outside of LIBERO, with no policy involved) by loading it into a minimal robosuite scene and sweeping it through an open-close cycle. This confirmed that the model compiles without mesh or collision errors and that the two sensor bodies are positioned symmetrically, facing one another, without interpenetrating the fingers, before any conclusions were drawn from downstream task performance.

With this validated gripper substituted for the stock one and the frozen OpenVLA-OFT policy otherwise unchanged, the full 500-episode evaluation protocol of Section 6.2 was repeated. Overall success rate dropped from the 98.2% baseline to **56.8**% ($284/500$), a 41.4-percentage-point reduction. Table 6.1 reports the per-task breakdown against the baseline.

Table 6.1: Per-task success rate on `libero_object`: stock Panda gripper (baseline) vs. the same policy with the DIGIT-equipped gripper substituted in, both evaluated over 50 trials per task.

Task	**Baseline**	**DIGIT gripper**	**Δ**
alphabet soup	1.00	0.32	-0.68
cream cheese	0.98	0.02	-0.96
salad dressing	0.96	1.00	+0.04
bbq sauce	1.00	0.42	-0.58
ketchup	1.00	1.00	0.00
tomato sauce	0.98	0.54	0.44
butter	1.00	0.00	-1.00
milk	0.98	0.66	-0.32
chocolate pudding	0.92	0.74	-0.18
orange juice	1.00	0.98	-0.02
Overall	**0.982**	**0.568**	**-0.414**

The drop is not uniform across tasks and the pattern is informative rather than merely discouraging. Three tasks (*ketchup*, *salad dressing*, *orange juice*) are essentially unaffected, while others (*butter*, *cream cheese*, *bbq sauce*, *alphabet soup*) collapse almost entirely. Manual inspection of the recorded rollout videos confirms the underlying cause: the two DIGIT sensor housings add physical thickness to the inner face of each finger, which changes the effective geometry of the grasp. For narrower objects, this routinely results in one of two failure modes visible directly in the rollouts: the gripper either fails to close far enough around the object to secure a grip or the object is effectively too wide for the now-narrower usable aperture between the two sensor housings and is pushed away rather than grasped. This is a mechanical, geometry-driven effect of the gripper modification itself, not a perception or policy failure: the frozen OpenVLA-OFT policy still commands grasps in essentially the same way it does with the stock gripper, but the physical hardware it is commanding has changed underneath it in a way the policy was never trained to account for.

This result is an expected and, in fact, informative consequence of the experimental design, rather than a flaw in it. OpenVLA-OFT's action head was trained and fine-tuned exclusively with the stock, sensor-free Panda gripper; it has no way of anticipating the small but task-relevant change in usable finger clearance introduced by the DIGIT housings. Because the policy remains entirely frozen throughout this thesis (Section 6.2), no mechanism exists yet for it to compensate for this shift. Recovering the lost success rate is precisely the role intended for the tactile-to-visual alignment module f_θ investigated in Chapter 5

and developed in Chapter 7: by construction, f_θ does not change the gripper geometry, but a well-aligned tactile signal is what would let a future, tactile-aware policy detect and correct for exactly this kind of grasp misjudgement in closed loop, rather than executing an open-loop grasp that was calibrated for a different hardware geometry. The 56.8% figure reported here is therefore best read as the frozen-policy, geometry-only baseline that any future tactile-aware policy on this hardware would need to improve upon, not as a final verdict on the viability of the platform.

6.5 Software Environment and Reproducibility

All experiments in this chapter were run on a Lightning.ai Studio instance with a single NVIDIA L4 GPU (24 GB VRAM), using `bfloat16` precision for OpenVLA-OFT inference. The core software stack consists of Python 3.10, PyTorch, Hugging Face Transformers, TensorFlow (required by LIBERO's benchmark utilities), MuJoCo, robosuite and the LIBERO and TACTO packages themselves. The evaluation code is the official `openvla-oft` repository, modified only to (i) register the custom `PandaDigitGripper` model described in Section 6.4 and (ii) select it via the `gripper_types` argument used when constructing the LIBERO environment; no other part of the evaluation pipeline, model loading or action-processing logic was altered.

7

Proposed Architecture

This chapter presents the architecture designed to test the central hypothesis of this thesis (Chapter 4): that a lightweight, trainable module can project tactile signals onto the visual latent space of a frozen Vision-Language-Action model, allowing the tactile representation to substitute for the visual one when vision is degraded or absent. The chapter also reports a pilot training study conducted on the $N = 80$ preliminary dataset (Chapter 5), which stress-tests the architecture end-to-end before it is deployed at scale on the LIBERO-based benchmark described in Chapter 6.

7.1 Overview and Design Principles

Given a tactile observation and a visual observation, the system produces a single embedding z that is fed to the frozen VLA in place of its standard visual embedding. Three design principles, established in the preceding chapters, constrain every architectural choice made in this chapter.

Cross-modal substitution. Unlike VLA-Touch [27] and Tactile-VLA [4], which add tactile feedback *alongside* an always-present visual stream, this architecture is designed around the scenario in which the visual stream becomes unreliable and must be *replaced*. The tactile-derived embedding is not an auxiliary signal to be blended with vision at every timestep; it is a candidate occupant of the same representational slot vision normally fills.

Frozen backbone. Consistent with VLA-Touch [27] and with the design principle already adopted for OpenVLA-OFT in Section 2.6, the VLA's vision encoder, language backbone and action head are never modified. All learnable capacity is concentrated in two lightweight external modules: the tactile encoder's channel adapter and the projection head f_θ.

Feasibility-first, data-driven design. Every component described below was chosen or revised on the basis of empirical evidence gathered in this thesis, rather than by default convention. Most notably, the tactile backbone selected in Section 7.2 deliberately departs from the obvious choice (the official TVL encoder) because the preliminary study in Chapter 5 showed that choice to be unsuitable on the data available to this project.

Figure 7.1 summarises the resulting pipeline: a frozen tactile encoder produces $z_{\mathcal{T}}$, a trainable projection head maps it to $\hat{z}_V$ in the visual latent space and a substitution mechanism decides, as a function of visual degradation, whether $\hat{z}_V$ or the native visual embedding $z_{\mathcal{V}}$ is passed downstream to the frozen VLA.

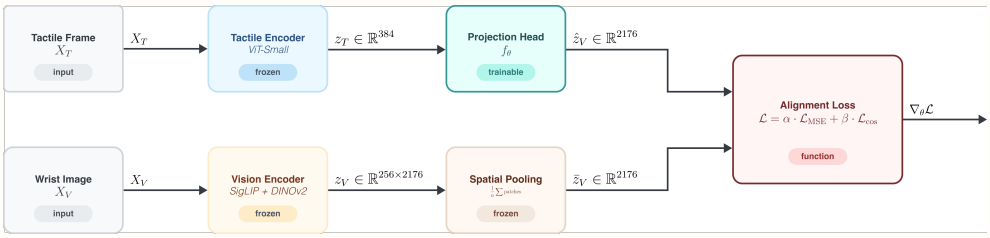


Figure 7.1: Overview of the proposed tactile-to-visual projection architecture. Frozen components are indicated; the channel adapter and the projection head f_θ are the only trainable modules.

7.2 Tactile Encoder

The tactile encoder maps a pair of raw DIGIT frames (one per gripper finger) into a single tactile descriptor $z_\mathcal{T} \in \mathbb{R}^{384}$. Two design decisions depart from the most obvious default and are both justified by evidence gathered earlier in this thesis rather than by convention.

Backbone choice: generic ViT-Small, not the official TVL encoder. The natural first candidate for a frozen tactile backbone is the official pre-trained encoder released with TVL [5], since it is trained specifically to produce tactile embeddings aligned with a visual-linguistic latent space. The preliminary study in Section 5.4, however, found that on the TACTO-simulated DIGIT frames used throughout this thesis, the official TVL encoder collapses into a near-degenerate, anisotropic region of its embedding space (mean inter-class cosine similarity 0.973, RSA correlation with the visual space $\rho = 0.372$), making it a poor regression source for f_θ. A generic ViT-Small backbone, pre-trained on ImageNet only and with no exposure to tactile data of any kind, was instead found to preserve a relational structure that correlates markedly better with the visual space ($\rho = 0.628$). This thesis therefore adopts the generic ViT-Small backbone as the frozen tactile encoder, a choice that is explicitly data-driven and provisional rather than a fixed design commitment (see the discussion in Section 5.5).

Dual-finger input: channel concatenation with a trainable adapter. The two DIGIT frames (left and right finger) are concatenated channel-wise into a single $(B, 6, H, W)$ tensor, rather than being processed independently and merged after encoding. A lightweight, trainable 1×1 convolution (`channel_adapter`, $6\to3$ channels, 18 parameters) maps this six-channel input down to the three-channel distribution expected by the frozen ViT and is initialised to compute the elementwise average of the two fingers, giving the network a sensible starting point before training begins. This adapter is the only trainable component inside the tactile encoder; the ViT backbone itself is kept entirely frozen (`requires_grad=False`), consistent with the frozen-backbone principle stated in Section 7.1.

7.3 Tactile-to-Visual Projection Head

The projection head f_θ is the only module in the entire pipeline responsible for bridging the modality gap characterised in Chapter 5:

$$\hat{z}_V = f_\theta(z_\mathcal{T}), \qquad f_\theta : \mathbb{R}^{384} \to \mathbb{R}^{2176}. \tag{7.1}$$

The target dimensionality, $d_V = 2176$, is fixed by OpenVLA-OFT's fused SigLIP–DINOv2 visual encoder (Section 2.5) and is not a free design choice. The head itself is implemented as a three-layer MLP with LayerNorm and GELU activations between layers and an unconstrained final linear projection (no output activation), since the target space $z_\mathcal{V}$ consists of unbounded real-valued coordinates (Section 5.3):

$$\text{Lin}(384 \to d_h) \to \text{LayerNorm} \to \text{GELU} \to \text{Lin}(d_h \to d_h) \to \text{LayerNorm} \to \text{GELU} \to \text{Lin}(d_h \to 2176). \tag{7.2}$$

LayerNorm is preferred over BatchNorm because it normalises along the feature dimension independently per sample, avoiding the batch-level variance estimation that becomes unstable at the small batch sizes used in this thesis's pilot experiments (Section 7.8); it is also the normalisation scheme used internally by the Llama 2 backbone inside OpenVLA-OFT. All linear layers are initialised with Xavier-uniform weights and zero biases.

The hidden width d_h is not fixed a priori: Section 7.8 reports a controlled comparison between $d_h = 1024$ (approximately 3.68×10^6 trainable parameters) and a substantially narrower $d_h = 256$ with dropout (approximately 2.4×10^5 trainable parameters, a $15\times$ reduction), motivated by a capacity-related failure mode observed during the pilot study.

7.4 Substitution Mechanism under Visual Degradation

The mechanism that decides whether the frozen VLA receives the native visual embedding or its tactile-derived counterpart is parameterised by a scalar degradation level rather than by an online, learned estimate of visual quality:

$$z = \alpha \cdot \hat{z}_V + (1 - \alpha) \cdot z_\mathcal{V}, \qquad \alpha \in [0, 1], \tag{7.3}$$

where α is assigned according to the known corruption level applied to the visual observation during the degradation curriculum (Section 6), rather than estimated from the corrupted image itself. This is a deliberate simplification relative to a fully autonomous quality-estimation module: because the severity of visual degradation is controlled experimentally in this thesis, α can be set directly from the known corruption level, isolating the question of *whether* the tactile-derived embedding can substitute for vision from the separate and harder question of *detecting*, from the visual signal alone, that substitution is needed. At $\alpha = 0$ the system reduces exactly to the unmodified frozen VLA; at $\alpha = 1$, corresponding to full visual blackout, action generation is driven entirely by $\hat{z}_V$. Replacing

this controlled schedule with a learned, image-conditioned estimator of α is identified as a direct extension for future work (Chapter 9), once the underlying projection f_θ has been validated at the scale required for reliable training (Section 7.8).

7.5 Integration with the Frozen VLA

The fused embedding z replaces the native visual embedding at the exact point in the OpenVLA-OFT pipeline where the fused SigLIP–DINOv2 features would otherwise be consumed: immediately before the two-layer MLP projector that maps visual features into the language model's embedding space (Section 2.5). No other part of the pipeline is touched. The language instruction is processed normally by the frozen Llama 2 backbone and the OFT-specific continuous action head (Section 2.6) generates actions conditioned on z and the language embedding exactly as it would from a clean visual observation. This integration point is what makes the substitution mechanism (Section 7.4) architecturally "invisible" to the rest of the frozen policy: from the perspective of the MLP projector and everything downstream of it, z is indistinguishable in shape and role from a native visual embedding, regardless of which branch produced it.

7.6 Training Objectives

f_θ is trained with a composite feature-alignment loss combining a magnitude term and a directional term:

$$\mathcal{L}_{\text{feat}} = \text{MSE}(\hat{z}_V, z_\mathcal{V}) + \lambda_{\cos}\big(1 - \cos(\hat{z}_V, z_\mathcal{V})\big), \tag{7.4}$$

with $\lambda_{\cos} = 1.0$, giving equal nominal weight to both terms. The MSE term constrains the absolute scale of the predicted embedding, while the cosine term constrains its angular orientation within the 2176-dimensional latent space, independently of magnitude. This combination is directly motivated by the cosine-similarity-based alignment mechanism validated in Section 3.6, where a comparable cosine objective against a frozen teacher representation was shown to mitigate representation degradation in a VLA's visual pathway; here, the same geometric principle is applied in the opposite direction, aligning a projected *tactile* representation onto a frozen *visual* target rather than realigning a fine-tuned visual representation onto its own frozen origin. It is also the same metric already used to characterise the modality gap empirically in Chapter 5, so that the training objective and the feasibility analysis that motivated it are measured on a consistent basis. Four complementary diagnostics are logged alongside the loss during training and validation: mean cosine similarity, its standard deviation across the batch, MSE and relative norm error $\big(\|\hat{z}_V\| - \|z_\mathcal{V}\|\big)/\|z_\mathcal{V}\|$, the last of which specifically isolates whether the projection head reproduces the correct *scale* of the target embedding, a property the cosine term alone cannot enforce.

7.7 Architectural Sanity Verification

Before any training on real data, the implementation was validated on synthetic tensors via five deterministic checks: (1) output dimensionality matches the target $(B, 2176)$; (2) the composite loss is finite (no `NaN`/`Inf`); (3) gradients reach both the channel adapter and the projection head; (4) the frozen ViT backbone receives no gradient; and (5) an optimiser step measurably updates every trainable parameter.

7.8 Training Strategy and Pilot Evaluation

With the architecture verified structurally, a pilot training study was conducted on the full $N = 80$ preliminary dataset (Section 5.1) to test the pipeline end-to-end before committing to full-scale training on LIBERO (Chapter 6). Two questions were addressed in sequence: whether the optimisation itself is well-behaved and, more critically, whether f_θ learns a mapping that depends meaningfully on the tactile input, as opposed to a shortcut that ignores it.

Optimisation behaviour: random per-pose split. f_θ ($d_h = 1024$) was first trained with a standard 90/10 random split over all 80 samples (AdamW, learning rate 10^{-4}, weight decay 10^{-4}, batch size 8, 50 epochs). Training converged cleanly and monotonically: validation cosine similarity rose from 0.488 at epoch 1 to 0.9975 at epoch 50, with validation MSE falling from 3.38 to 0.024 and no instability at any point (Figure 7.2). Because this split assigns different grasp poses of the *same* object to training and validation, it establishes only that the optimisation procedure is numerically well-behaved; it does not test generalisation to unseen objects, since the visual target space already exhibits near-identical intra-class embeddings across poses of the same object (Section 5.3).

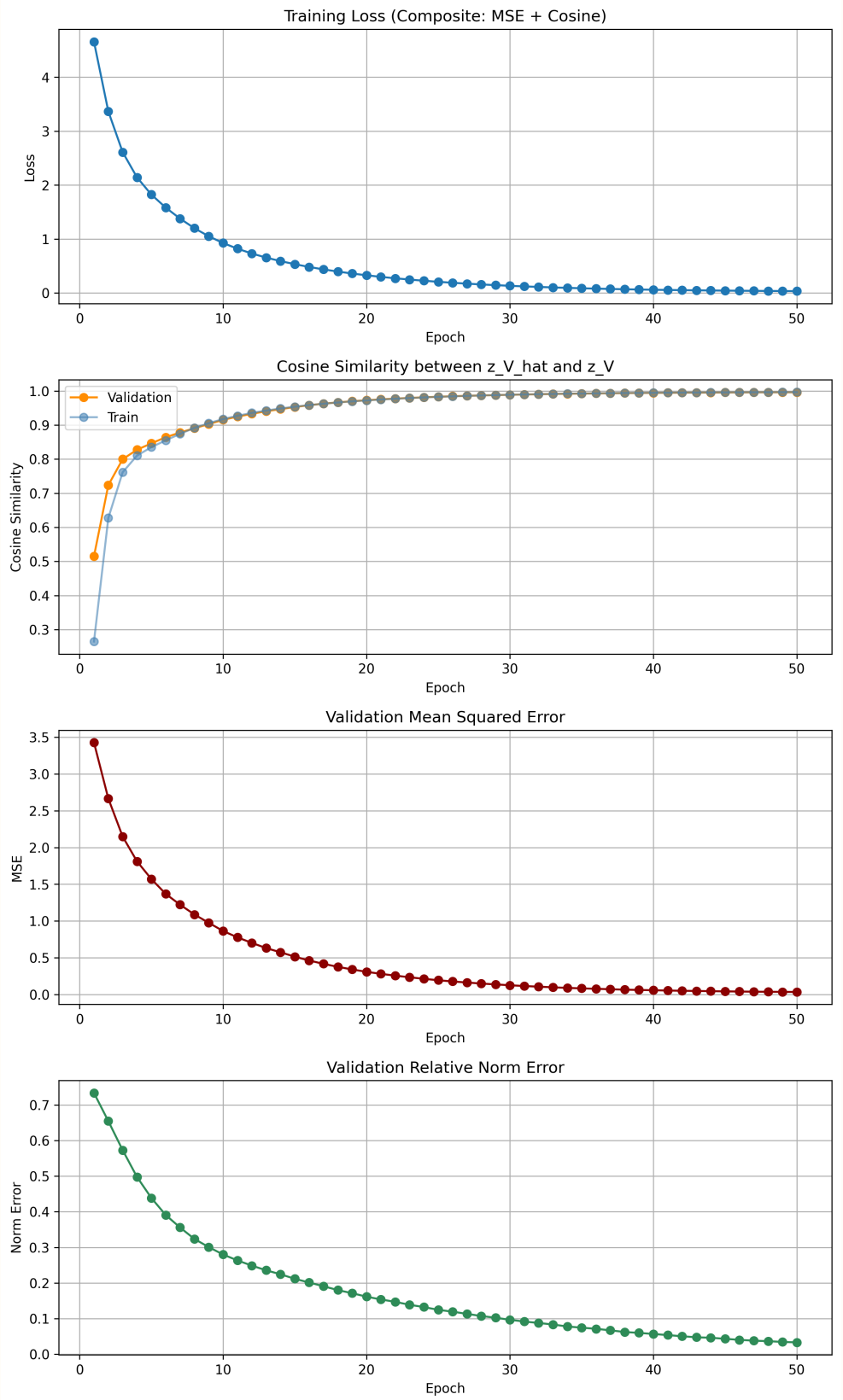


Figure 7.2: Pilot training on the random per-pose split ($d_h = 1024$): composite loss, cosine similarity (train and validation), validation MSE and validation relative norm error over 50 epochs. Convergence is smooth and monotonic, confirming the pipeline is numerically well-behaved, but the near-perfect final cosine similarity reflects generalisation across poses of already-seen objects rather than across unseen objects (overfitting).

Cross-object generalisation: leave-one-object-out. To test generalisation to genuinely unseen objects, f_θ was retrained from scratch in a leave-one-object-out (LOO) protocol: for each of the eight objects in turn, all samples of the remaining seven objects were used for training (70 samples) and every pose of the held-out object was used exclusively for validation (10 samples), never seen during training. Table 7.1 reports the final validation cosine similarity for each fold, alongside a trivial constant-vector baseline that ignores the tactile input entirely and always predicts the mean of the training-set visual embeddings.

Table 7.1: Leave-one-object-out cross-validation: final validation cosine similarity of f_θ ($d_h = 1024$) against a naive constant-vector baseline, per held-out object.

Held-out object	f_θ **CosSim**	**Baseline CosSim**	**Margin**
abstract_ball	0.8228	0.8406	-0.0178
cage	0.7113	0.7072	$+0.0040$
cube	0.8777	0.8806	-0.0029
decorative_ball	0.9302	0.9032	$+0.0270$
disco_ball	0.9322	0.9086	$+0.0236$
greek_dice	0.7868	0.8260	-0.0392
sphere	0.9181	0.9264	-0.0083
wood_dice	0.8360	0.8550	-0.0190
Mean	**0.8519** $\pm$ 0.0781	**0.8559**	-0.0041

The mean cross-object cosine similarity of f_θ (0.852) is, on average, statistically indistinguishable from the naive baseline (0.856): the mean margin is slightly negative and the sign of the margin is inconsistent across folds, positive on three objects and negative on five. This is a materially different, and considerably more sobering, result than the near-perfect cosine similarity obtained under the random per-pose split. The per-fold pattern is nonetheless not random: the two folds with the lowest absolute performance for both f_θ and the baseline are `cage` and `greek_dice`, the two objects already flagged in Chapter 5 as having the least distinctive tactile and visual signatures relative to the rest of the object set, while the three visually near-identical smooth spheres (`decorative_ball`, `disco_ball`, `sphere`) achieve the highest absolute scores for both methods: consistent with the visual embedding space's high inter-class similarity for these three objects (Section 5.3) making the regression target easier to approximate regardless of the tactile input.

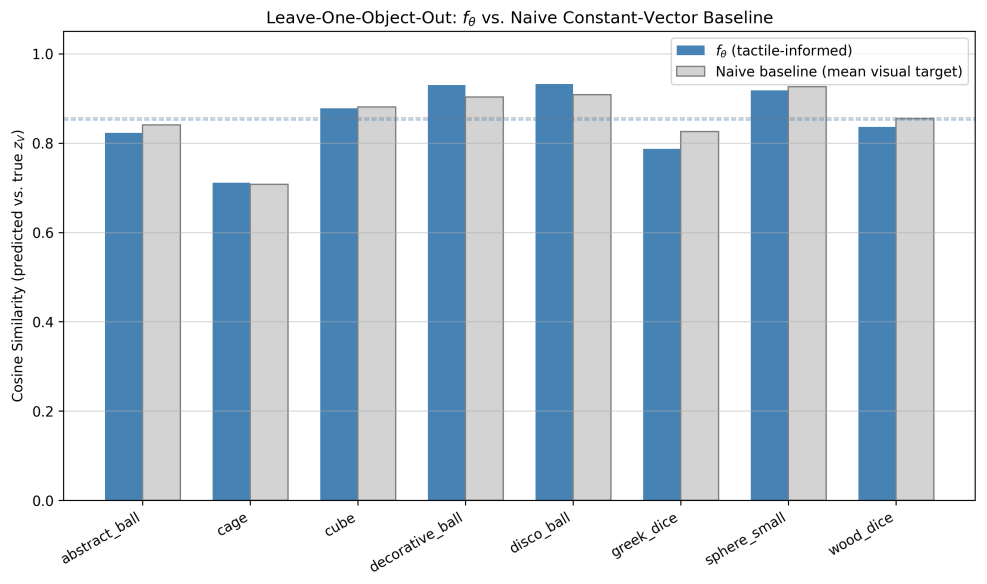


Figure 7.3: Leave-one-object-out validation cosine similarity: f_θ ($d_h = 1024$) versus the naive constant-vector baseline, per held-out object. Dashed lines mark the respective means. The two methods are close on every fold, with no consistent advantage for f_θ.

A capacity-reduction attempt and why it did not help. The near-tie with the naive baseline raised a specific hypothesis: with only 70 training samples per fold and approximately 3.68×10^6 trainable parameters in the $d_h{=}1024$ head, f_θ may have sufficient capacity to converge toward a solution close to the training-set mean, regardless of the tactile input, without being forced by the optimisation to actually use that input. To test this, the LOO protocol was repeated with a substantially narrower head ($d_h{=}256$, approximately 2.4×10^5 parameters, a $15\times$ reduction) and dropout ($p{=}0.3$) added after each hidden layer, together with a stronger weight decay (10^{-2}, up from 10^{-4}).

Table 7.2 reports the outcome. Contrary to the capacity-collapse hypothesis, reducing the head's capacity did not close the gap to the baseline instead it *widened* it, uniformly and substantially: every one of the eight folds now shows a negative margin and the mean margin worsens from -0.004 ($d_h{=}1024$) to -0.072 ($d_h{=}256$).

Table 7.2: Leave-one-object-out cross-validation with a narrower projection head ($d_h{=}256$, dropout 0.3, weight decay 10^{-2}), against the same naive baseline as Table 7.1.

Held-out object	f_θ **CosSim** ($d_h{=}256$)	**Baseline CosSim**	**Margin**
abstract_ball	0.7420	0.8406	-0.0986
cage	0.6124	0.7072	-0.0948
cube	0.8000	0.8806	-0.0806
decorative_ball	0.8532	0.9032	-0.0500
disco_ball	0.8695	0.9086	-0.0391
greek_dice	0.7685	0.8260	-0.0575
sphere	0.8562	0.9264	-0.0702
wood_dice	0.7704	0.8550	-0.0846
Mean	**0.7840** $\pm$ 0.0838	**0.8559**	-0.0719

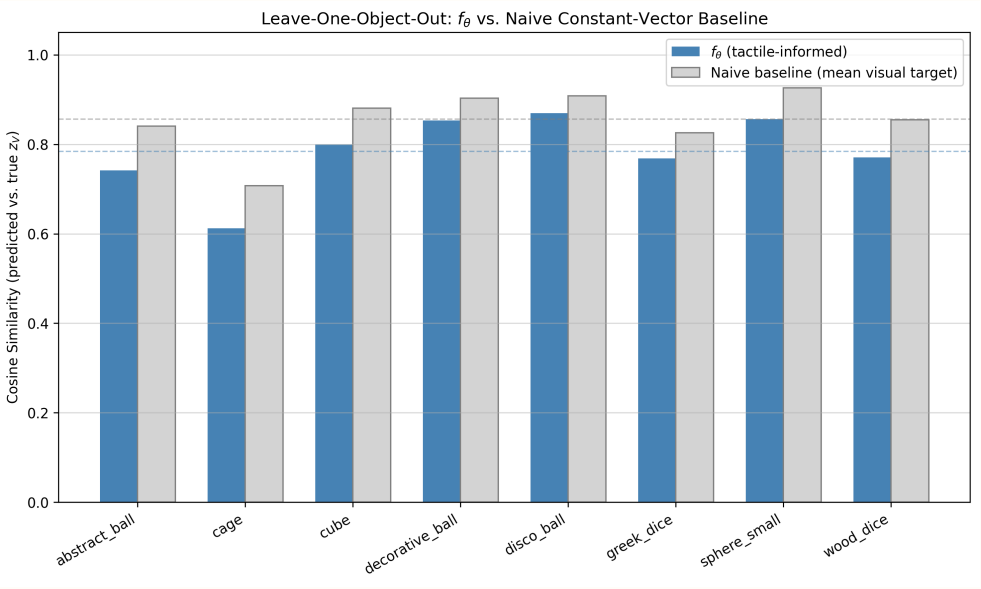


Figure 7.4: Leave-one-object-out validation cosine similarity with the narrower head ($d_h{=}256$): f_θ versus the naive baseline, per held-out object. Unlike Figure 7.3, the baseline now dominates on every single fold.

This outcome is informative precisely because it rules out the simplest explanation. If the original head's capacity had been the limiting factor, constraining it should have forced the network to rely more on $z_\mathcal{T}$ and narrowed the gap to the baseline; instead, absolute performance dropped on every fold and the gap widened uniformly. This pattern is more consistent with an under-fitting regime induced by the combination of a smaller network, stronger regularisation, and very few training samples, than with the original network having been forced into a lazy, input-independent solution. It does not, by itself, distinguish

between two remaining explanations: that 70 training samples are simply insufficient for any head of this general design to extract a reliable signal from $z_{\mathcal{T}}$ or that the tactile representation $z_{\mathcal{T}}$ itself (produced by a generic, non-tactile-specialised ViT-Small backbone (Section 7.2))does not carry enough object-discriminative information beyond what the training-set mean already captures, at this dataset scale.

Summary and implications. Three findings emerge from this pilot study. First, the training pipeline is numerically correct and stable: gradients flow to the intended parameters only, the frozen backbones remain untouched and optimisation converges smoothly under both configurations tested (Section 7.7, Figure 7.2). Second, on the current $N = 80$ dataset, f_θ has not yet been shown to extract cross-object generalisable information from the tactile signal beyond what a trivial, tactile-blind baseline already achieves; this held under two substantially different head capacities, suggesting the limitation is more likely attributable to dataset scale than to this specific architectural choice, though the evidence gathered here cannot fully disambiguate the two. Third, and consequently, the numerical results in this section should be read strictly as a pilot feasibility and debugging exercise, not as a validation of the projection module's ability to generalise. Resolving this open question is the direct object of the full-scale training enabled by the LIBERO-based benchmark of Chapter 6, whose substantially larger and more diverse dataset is the most direct lever available for distinguishing a data-scale limitation from a representational one.

7.9 Comparison with Prior Work

Table 7.3 situates the proposed architecture relative to the tactile-VLA and cross-modal alignment literature reviewed in Chapter 3, along the dimensions most relevant to the design choices made in this chapter.

Table 7.3: Architectural comparison with the most closely related prior work. ✓ = present, × = absent.

	TVL	**VLA-Touch**	**Tactile-VLA**	**This work**
Frozen VLA backbone	n/a	✓	partial	✓
Optical tactile images	✓	×	×	✓
Explicit latent substitution	×	×	×	✓
Degrades to base VLA	—	—	—	✓
Robotics downstream task	×	✓	✓	✓
Cross-object validation reported	×	×	×	✓

The last row is deliberately marked as inconclusive rather than as a checkmark: to the

best of the review conducted in Chapter 3, none of the compared works report a leave-one-object-out–style evaluation of their tactile representation against a trivial baseline, making the negative result of Section 7.8 a methodological contribution in its own right, independent of whether the underlying projection module is ultimately shown to work at scale.

8

Discussion

The preceding chapters establish a chain of evidence rather than a single result: a feasibility analysis at the level of embeddings (Chapter 5), a working but mechanically imperfect benchmark at the level of a real manipulation policy (Chapter 6) and a pilot training study at the level of the projection module itself (Chapter 7).

8.1 A Necessary but Not Sufficient Condition

Chapter 5 showed that a generic ViT-Small tactile encoder produces embeddings whose relational structure correlates positively with that of the frozen visual space ($\rho = 0.628$) and on that basis this thesis concluded that cross-modal alignment was, in principle, *feasible*. The pilot study of Section 7.8 does not contradict that conclusion, but it does sharpen it considerably: representational compatibility at the level of aggregate, class-wise similarity structure is a necessary condition for a learned projection to succeed, not a sufficient one. The RSA correlation in Chapter 5 was computed over similarities between object-level clusters, using all eighty samples; it says that the two modalities broadly agree on *which objects are similar to which*. Learning f_θ to generalise to a held-out object, by contrast, requires inferring a rule general enough to extrapolate from seven object classes to an eighth never seen in any form during training: a substantially harder demand than the one the feasibility study actually tested. In hindsight, the gap between these two notions of ”the alignment is feasible” was under-examined in Chapter 5; the pilot study in Section 7.8 was precisely what surfaced it, and did so through a methodologically appropriate comparison (leave-one-object-out against a trivial baseline) rather than through an unqualified success metric that would have obscured the distinction.

8.2 Disambiguating Data Scale from Representational Capacity

Section 7.8 deliberately tested whether f_θ’s capacity, rather than the tactile representation itself, was responsible for the tie with the naive baseline, by retraining with a $15\times$ narrower head. The result (a uniform *worsening* of the margin against the baseline, not an improvement) rules out the simplest capacity-based explanation, but it does not, by itself, distinguish between the two explanations that remain: that 70 training samples per fold are simply too few for any projection head to extract a reliable cross-object signal or that the tactile representation $z_\mathcal{T}$ produced by a generic, non-tactile-specialised ViT-Small backbone does not carry enough object-discriminative information beyond the training-set

mean, at any sample size. These two explanations have very different implications and this thesis' evidence base does not adjudicate between them. What it does provide is context that makes the data-scale explanation more plausible than it might first appear: eight object classes is small even by the standards of few-shot learning and the leave-one-object-out protocol asks f_θ to generalise having seen literally zero examples of the held-out object's shape or texture, in either modality, during training. A representation could carry genuinely useful geometric information and still fail to support extrapolation this abrupt without considerably more training diversity than seven object classes provide.

8.3 An Independent Confound: The Gripper Itself

A separate and, in some ways, more immediately actionable finding comes from Chapter 6: substituting the stock Panda gripper for the DIGIT-equipped version drops the frozen policy's success rate from 98.2% to 56.8% *before any tactile alignment is involved at all*. This result is orthogonal to the central hypothesis of this thesis and is worth keeping conceptually separate from it. Even a perfectly generalising f_θ would not, by itself, recover this gap, because the gap is mechanical: the added thickness of the sensor housings changes the effective grasp aperture, not perceptual. The frozen VLA policy was fine-tuned on the stock gripper geometry and has no mechanism for adapting to a change in the hardware it is commanding, regardless of what signal occupies its visual embedding slot. This means that, in the architecture proposed in Chapter 7, tactile-to-visual substitution and gripper-geometry compensation are two distinct problems that this thesis' experimental design does not disentangle: the LIBERO benchmark, as instrumented in Chapter 6, measures both simultaneously and any future full-scale evaluation of f_θ in that environment will need to account for this confound explicitly (for instance, by fine-tuning the action head on the modified gripper's kinematics independently of tactile alignment) in order to isolate what f_θ itself contributes.

8.4 Revisiting the Choice of Tactile Backbone

Chapter 5 adopted a generic ViT-Small encoder over the official TVL encoder on the grounds that the latter's embeddings collapsed into a near-degenerate, anisotropic region of its embedding space on the TACTO-simulated data used throughout this thesis, and explicitly flagged a plausible alternative explanation: a sim-to-real domain gap, since TVL was trained exclusively on real DIGIT captures. The pilot results in Chapter 7 do not resolve this question, but they do raise its stakes. If the sim-to-real explanation is correct, then the generic ViT-Small backbone selected on data-driven grounds may itself be a source of the representational ceiling discussed in Section 8.2: an encoder with no exposure to tactile data of any kind, simulated or real, may simply lack the inductive bias needed to extract fine contact geometry efficiently from few examples, even where it is not anisotropically collapsed. This thesis cannot distinguish, with the evidence gathered, whether a better-

suited tactile backbone (e.g. TVL fine-tuned on simulated frames or a domain-randomised variant) would have narrowed the gap to the naive baseline in Section 7.8; it can only note that the backbone choice, correct on the grounds it was made, remains an open variable rather than a settled one.

8.5 Summary

Taken together, the evidence in this thesis supports a qualified rather than unqualified answer to the research question posed in Chapter 4. Tactile and visual embeddings occupy compatible enough regions of representational space, in aggregate, to motivate the pursuit of a learned projection (f_θ) between them; but at the data scale available to this thesis, that projection has not yet been shown to generalise to unseen objects beyond what a trivial, tactile-blind baseline already achieves, and a second, independent mechanical confound (the physical footprint of the tactile sensors themselves) degrades policy performance regardless of the alignment question entirely. Neither finding invalidates the thesis' central hypothesis; both narrow, considerably, the conditions under which it remains open to being confirmed.

9

Conclusion and Future Work

9.1 Summary of Contributions

This thesis set out to investigate whether a lightweight, trainable module could project tactile signals into the frozen visual latent space of a Vision-Language-Action model, allowing tactile input to substitute for degraded or absent vision without modifying the underlying policy. Rather than committing directly to full-scale training, the work adopted a feasibility-first methodology and its contributions follow that structure.

A feasibility analysis grounded in data, not assumption (Chapter 5). Before any architecture was trained, this thesis quantified whether a tactile and a visual embedding space are even compatible enough to be worth aligning, using representational similarity analysis on a small but carefully designed exploratory dataset. This analysis produced a concrete, evidence-based decision: adopting a generic ViT-Small tactile backbone over the official TVL encoder, together with an explicit, honestly stated alternative explanation (a sim-to-real domain gap) for why the more obvious choice underperformed.

A fully reproducible, physically instrumented benchmark (Chapter 6). This thesis integrated a custom DIGIT-equipped Panda gripper into the LIBERO/robosuite/MuJoCo simulation stack, bridging TACTO's tactile rendering pipeline (originally validated in a separate, PyBullet-based pipeline) into a physically different simulator and quantified, rather than assumed, the policy-level consequences of that hardware change: a drop from 98.2% to 56.8% success rate on `libero_object`, attributable to a specific, identified mechanical cause rather than left unexplained.

A rigorous pilot evaluation of the core hypothesis (Chapter 7). The projection module f_θ was implemented, verified to train in a numerically well-behaved way and evaluated under a leave-one-object-out protocol against a naive baseline: a comparison absent from every closely related prior work reviewed in Chapter 3 (Table 7.3). The resulting negative/inconclusive result is reported as such, together with a follow-up experiment (capacity reduction) designed to narrow down its cause, rather than reframed as a success.

An honest account of what remains open (Chapter 8), separating the central cross-modal alignment question from an independent mechanical confound and identifying the specific ambiguity, data scale versus representational capacity, that future work would need to resolve.

Taken as a whole, this thesis' contribution is best characterised as a validated experimental infrastructure and a disciplined feasibility study, rather than a demonstrated end-to-end capability: it establishes what is plausible, builds and documents what is

needed to test it properly and reports honestly where the current evidence stops.

9.2 Limitations

Three limitations, already surfaced individually in earlier chapters, are worth restating together, since they jointly bound how far the thesis' conclusions can be read.

First, **dataset scale**. Every embedding-level and training-level result in this thesis rests on eight object classes and eighty samples (Section 5.1). This is sufficient to probe gross distributional and relational properties of two embedding spaces and to run a pilot training loop end-to-end, but it is not sufficient to train or validate a projection module intended to generalise across novel objects, as Section 7.8 demonstrates directly.

Second, **simulator- and backbone-specificity**. The feasibility conclusions of Chapter 5, and the tactile backbone choice that followed from them, are specific to TACTO-simulated DIGIT frames and to a ImageNet-pretrained ViT-Small encoder; neither conclusion has been tested against real tactile hardware and Section 8.4 explicitly leaves open whether a different backbone would alter the pilot's outcome.

Third, **an unresolved mechanical confound**. The gripper modification required for tactile sensing degrades frozen-policy performance independently of any tactile alignment (Section 8.3) and this thesis' experimental design does not yet separate that effect from the effect of the projection module itself.

9.3 Future Work

The most direct and highest-priority next step is **scaling up data collection** using the benchmark infrastructure already built and validated in Chapter 6: substantially more objects, grasp poses and episodes than the eighty-sample pilot, collected directly within the LIBERO/TACTO-MuJoCo pipeline rather than in the standalone setup of Chapter 5. This is the most direct lever available for disambiguating the two explanations left open in Section 8.2: insufficient data versus insufficient tactile representation, since it directly tests the one variable (training set size and diversity) that the pilot study could not vary.

Second, **revisiting the tactile backbone** once more data is available, including candidates deliberately excluded or untested in this thesis: the official TVL encoder fine-tuned on TACTO-simulated frames (directly testing the sim-to-real hypothesis of Section 5.4), domain-randomised simulated tactile rendering or a tactile encoder trained jointly with f_θ rather than kept entirely frozen.

Third, **isolating the gripper confound** identified in Section 8.3, for instance by fine-tuning the frozen policy's action head on the DIGIT-equipped gripper's kinematics alone, without any tactile input, to establish a corrected baseline against which the contribution of f_θ specifically, rather than the contribution of the gripper redesign as a whole, can be measured.

Fourth, **implementing the visual degradation protocol** originally scoped as Section 6.5

of this thesis and deferred here in favour of prioritising the more foundational questions above. Systematically degrading or occluding the `agentview` input and measuring the substitution mechanism's behaviour under graded degradation remains necessary before the architecture's motivating scenario (a VLA operating under genuinely degraded vision) can be evaluated directly and is a natural companion to the larger-scale training proposed above.

Finally, once cross-object generalisation is demonstrated to exceed the naive baseline at a larger data scale, **policy-level evaluation of the full substitution mechanism** inside the corrected LIBERO benchmark would close the loop opened by this thesis: measuring whether a tactile-derived embedding can, in practice, recover manipulation success under visual degradation, rather than only approximate the frozen visual target in isolation.

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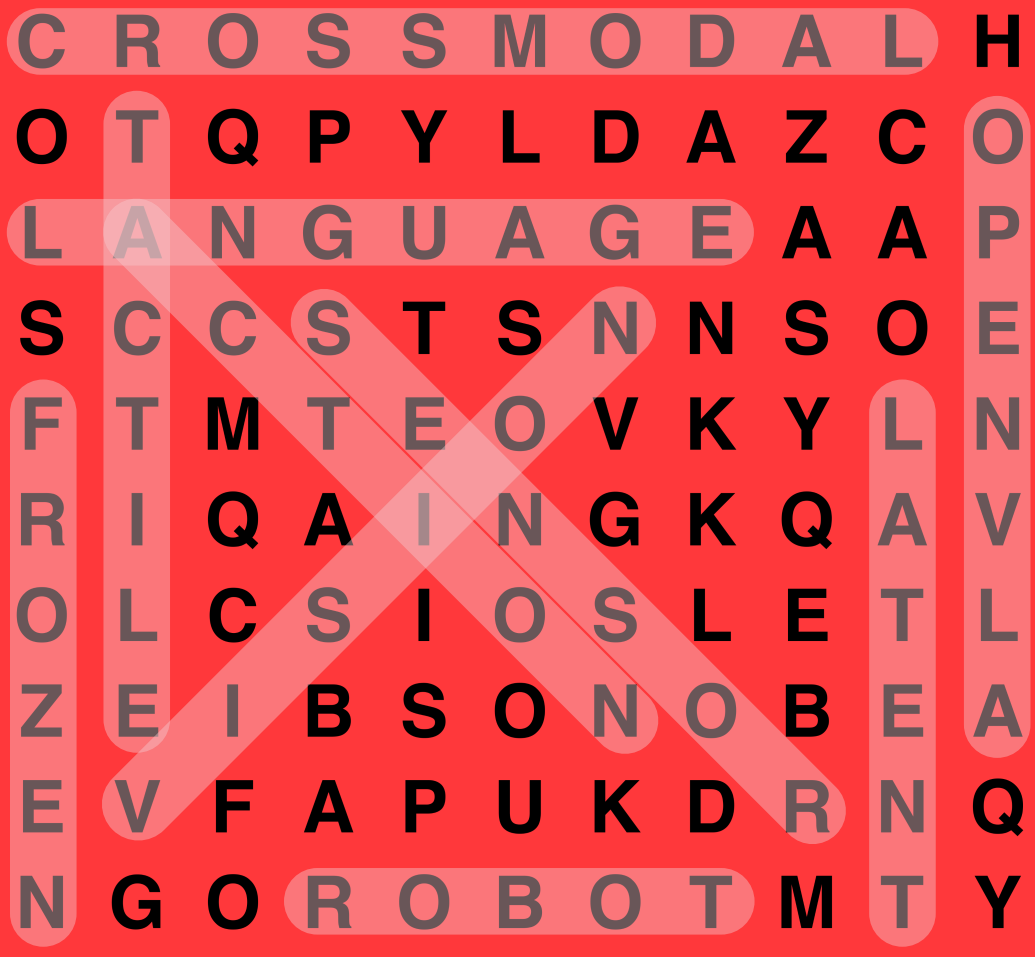
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Solution.



Cross-modal Tactile OpenVLA Frozen Latent Vision Action Language Sensor Robot